



WSQ · INSTRUCTOR-LED · 2 DAYS

Developing AI Apps and Agents on Azure (AI-103)

Course Code: TGS-2023036651

Tertiary Infotech Academy Pte Ltd
UEN 201200696W

Version 4.0 · 27 Sep 2026



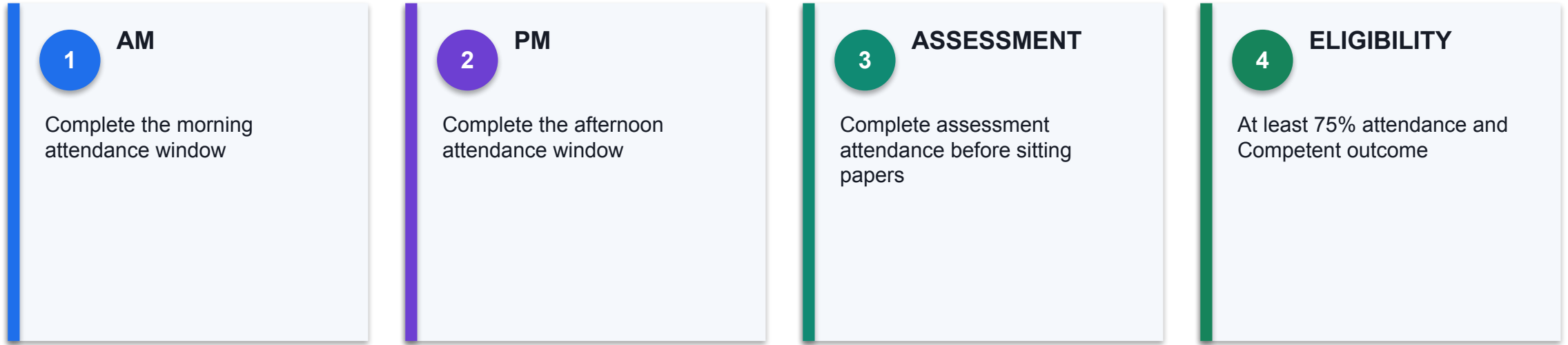
TOPIC 00

00

Course administration

WSQ requirements · learning outcomes · evidence expectations

Digital Attendance (Mandatory)



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

About the Trainer



Tertiary Infotech Academy Trainer

Microsoft Azure AI
training team

NAME	Tertiary Infotech Academy Trainer
TITLE / DESIGNATION	WSQ Technical Trainer
QUALIFICATIONS	Azure AI and adult-learning credentials
AREAS OF EXPERTISE	Microsoft Foundry, Azure AI services, MLOps
EXPERIENCE	Applied cloud AI delivery and facilitation
CONTACT	Introduced during the opening briefing

About the Trainer



Dr. Alfred Ang

Principal Trainer
AI, automation, and software engineering

NAME	Dr. Alfred Ang
TITLE / DESIGNATION	Principal Trainer
QUALIFICATIONS	PhD; Azure AI and software engineering
AREAS OF EXPERTISE	AI solutions, agents, automation, apps
EXPERIENCE	Industry implementation and adult learning
CONTACT	Shared during class

Ground Rules

1

PUNCTUAL

Return from breaks on time

2

PARTICIPATE

Explain decisions and evidence

3

PROTECT DATA

Use synthetic/non-confidential inputs

4

ASK EARLY

Raise blockers before they cascade

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Learning Outcomes

1

LO1

Analyse Azure AI algorithms and efficiency

2

LO2

Evaluate strengths and limitations

3

LO3

Assess feasibility and improvements

4

EVIDENCE

Explain what proves each conclusion

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Two-Day Lesson Plan

1

DAY 1 AM

Planning, governance,
generative AI

2

DAY 1 PM

Agents, vision,
language/speech

3

DAY 2 AM

Custom language, search,
extraction

4

DAY 2 PM

Capstone + 2-hour
assessment

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

AI-103 Course Scope

1

EXAM

AI-103: Developing AI Apps
and Agents on Azure

2

SCOPE

Skills measured from 16 Apr
2026

3

COURSE

WSQ TGS-2023036651 · two
days

4

FOCUS

Foundry apps, agents,
multimodal and extraction

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Current Microsoft certification pathway

AI-103: Developing AI Apps and Agents on Azure · skills measured from 16 Apr 2026

[Microsoft Learn · AI-103 Study Guide](#)

OPEN RESOURCE

Briefing for Assessment

1

OPEN BOOK

Use approved slides and
Learner Guide

2

INDIVIDUAL

No discussion or shared
answers

3

EVIDENCE

Answer every K/A criterion

4

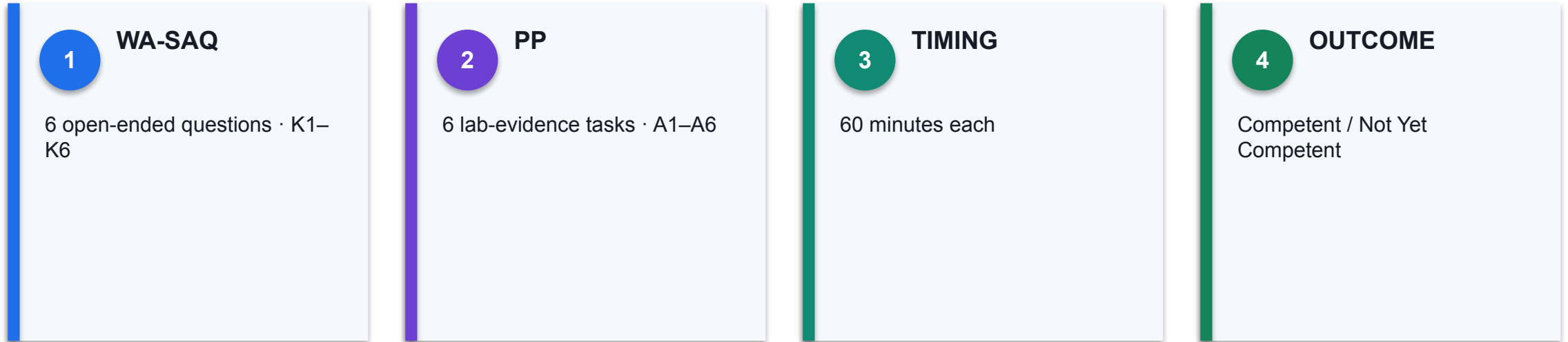
SUBMISSION

Upload through the LMS

ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

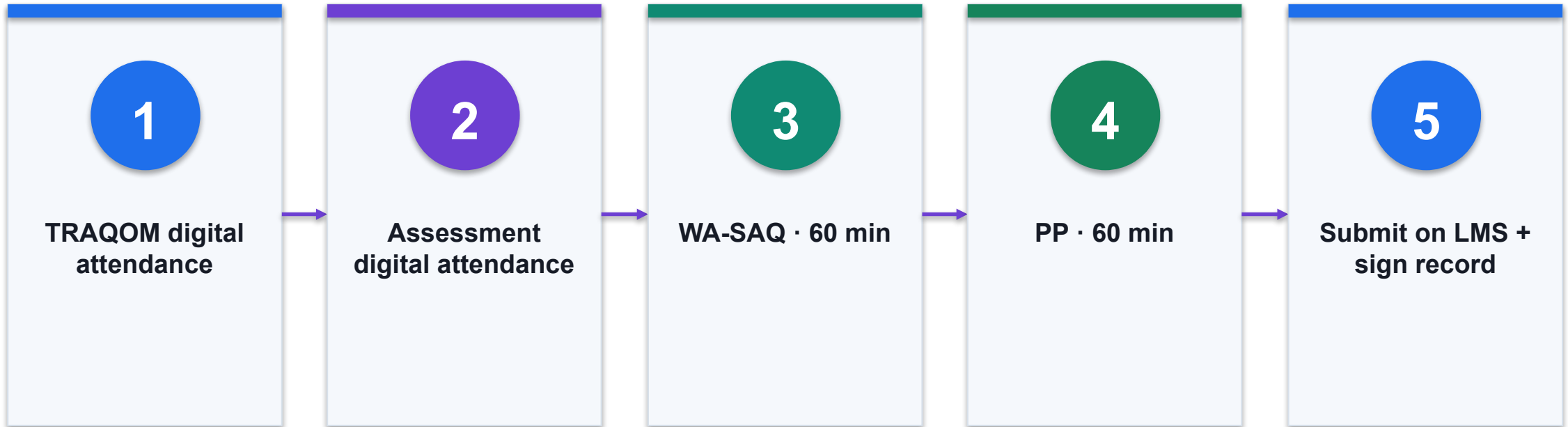
Assessment



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Assessment Flow



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Courseware and Assessment on the LMS

Download the learner materials and submit the two candidate papers on the LMS

[LMS-TMS · Courseware and Assessment](#)

OPEN RESOURCE

Access the Hands-On Labs

Clone the repository or use GitHub · Code · Download ZIP; every lab has its own folder and detailed README

[GitHub · AI-103 Hands-On Labs](#)

OPEN RESOURCE

TOPIC 00

00

Day 1

Planning · governance · generative AI · agents · vision · language

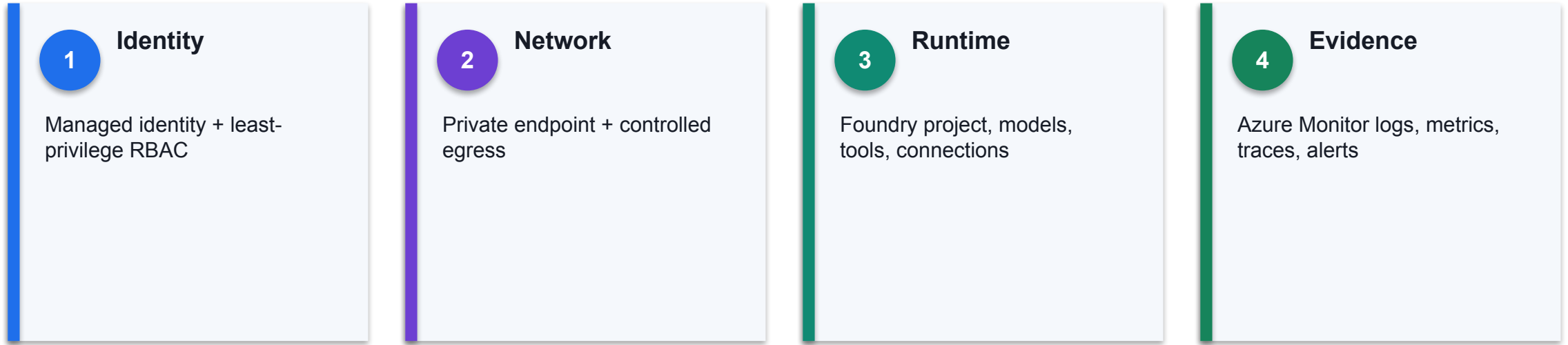
TOPIC 01

01

Plan, secure, and operate Foundry services

PLAN & MANAGE · AI-103 25–30%

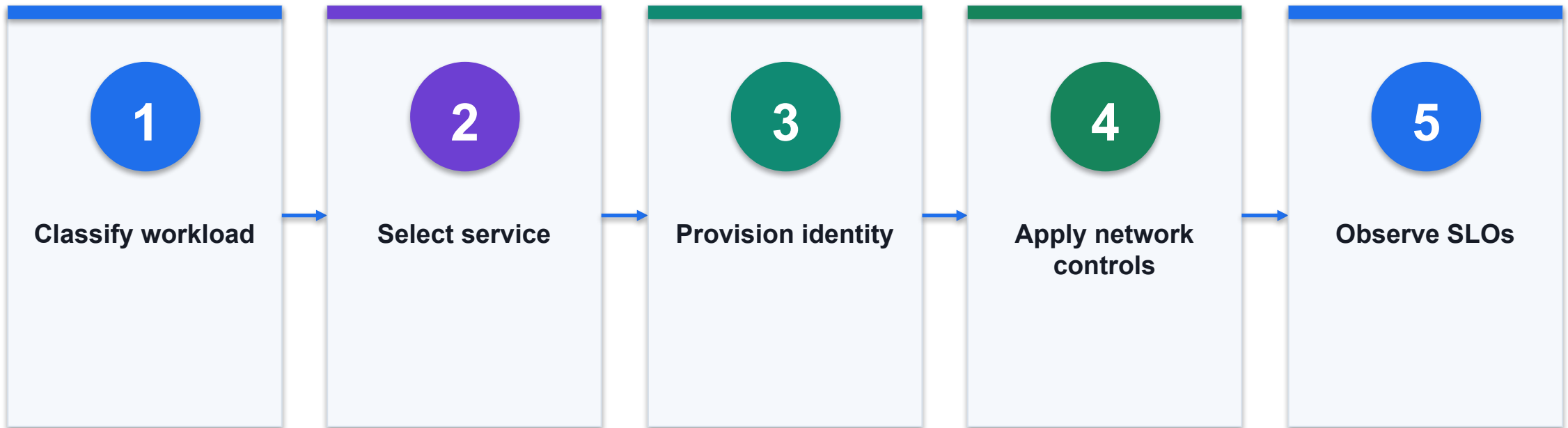
Plan, secure, and operate Foundry services · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Plan, secure, and operate Foundry services · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Plan, secure, and operate Foundry services · exact contract

Contract	Exact value / field	Engineering consequence
Project endpoint	https://<resource>.services.ai.azure.com/api/projects/<project>	Foundry SDK
Auth	DefaultAzureCredential	Entra token; avoid embedded keys
RBAC	Cognitive Services User	Data-plane inference
Diagnostics	RequestResponse + Audit	Send to Log Analytics

Plan, secure, and operate Foundry services · implementation excerpt

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
client = AIProjectClient(endpoint=PROJECT_ENDPOINT,

credential=DefaultAzureCredential())
for conn in client.connections.list():
    print(conn.name, conn.type)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

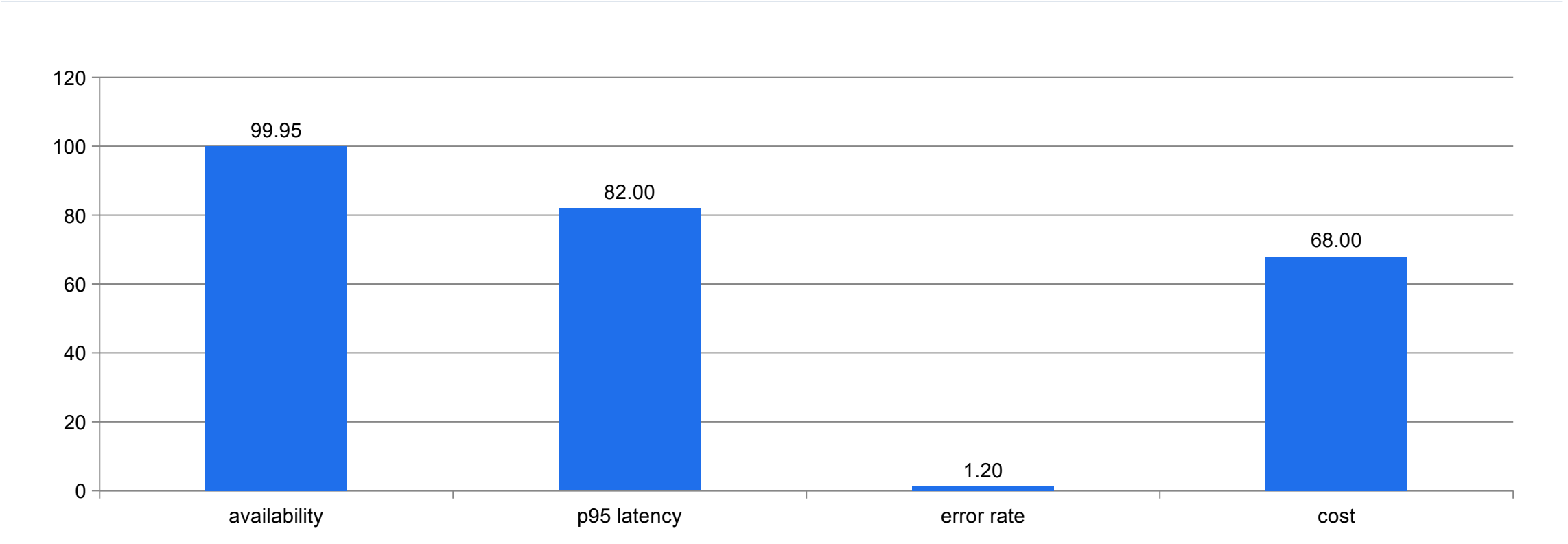
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

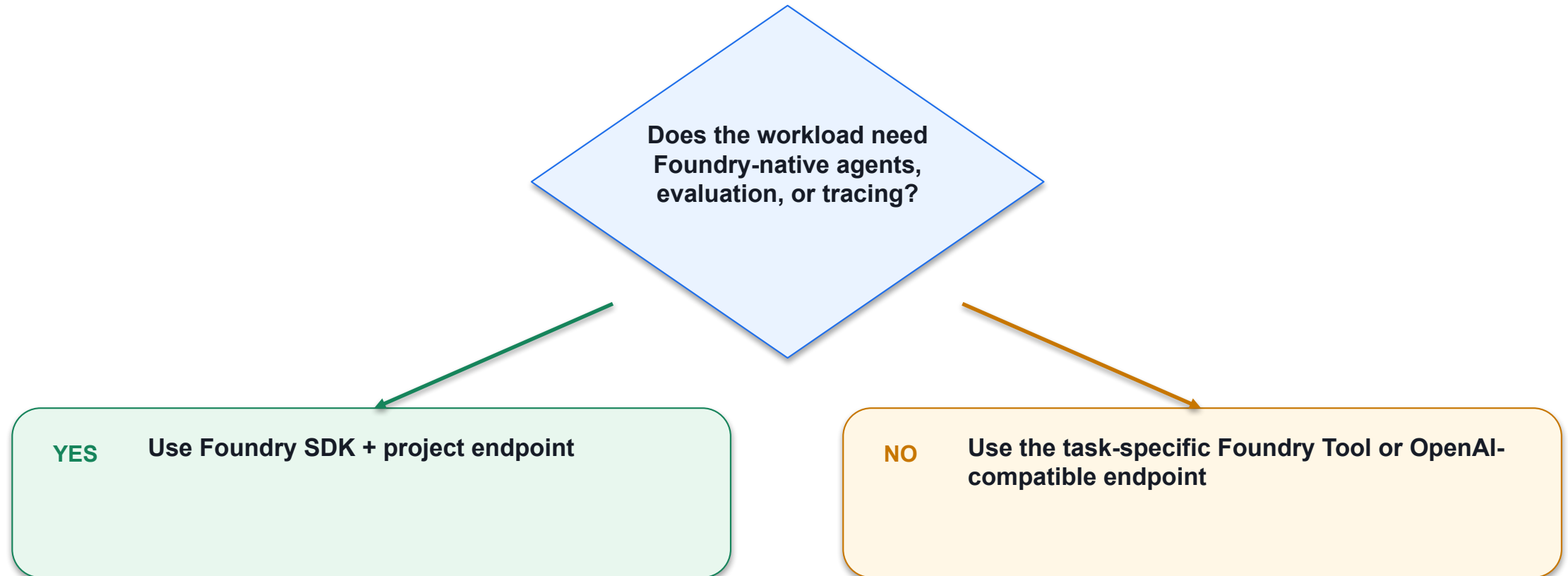
Handle auth, quota, timeout, and invalid-state paths.

Plan, secure, and operate Foundry services · engineering metric



WHAT THE DATA SHOWS Indicative dashboard: availability is a percentage; other values are normalized to target. Investigate any measure below its target band.

Plan, secure, and operate Foundry services · decision rule



Plan, secure, and operate Foundry services · failure and control

 401 / 403 LIKELY CAUSE Identity lacks data-plane role CONTROL / RECOVERY Verify principal, scope, and token audience	 429 LIKELY CAUSE Quota or rate limit CONTROL / RECOVERY Retry with backoff; request quota or scale units	 Timeout LIKELY CAUSE Network/DNS/private link CONTROL / RECOVERY Resolve FQDN from workload subnet	 Blind spot LIKELY CAUSE Diagnostics not routed CONTROL / RECOVERY Enable categories and alert ownership
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 01: Plan, Manage, Monitor, and Secure an Azure AI Solution

Mechanism focus: Plan, secure, and operate Foundry services



YOU WILL PRODUCE You should have a service map, security plan, monitoring plan, and architecture diagram.

Complete click-by-click procedure: [Learner Guide · Lab 01 · labs/lab-01-plan-manage-secure-azure-ai-solution/README.md](#)

Lab 01: acceptance evidence

PASS WHEN

Evidence pack: resource topology, RBAC assignment, private connectivity decision, successful SDK call, and an alert driven by a real metric.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 02

02

Responsible AI, content safety, and governance

PLAN & MANAGE · AI-103 25–30%

Responsible AI, content safety, and governance



Input guard

Prompt shields + content classification

Model policy

Deployment filters + approved model/version

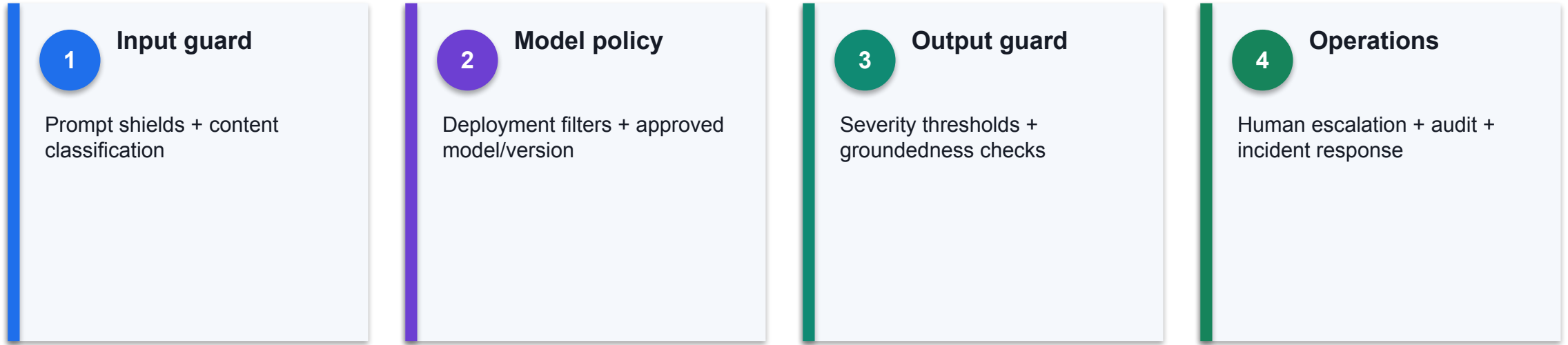
Output guard

Severity thresholds + groundedness checks

Operations

Human escalation + audit + incident response

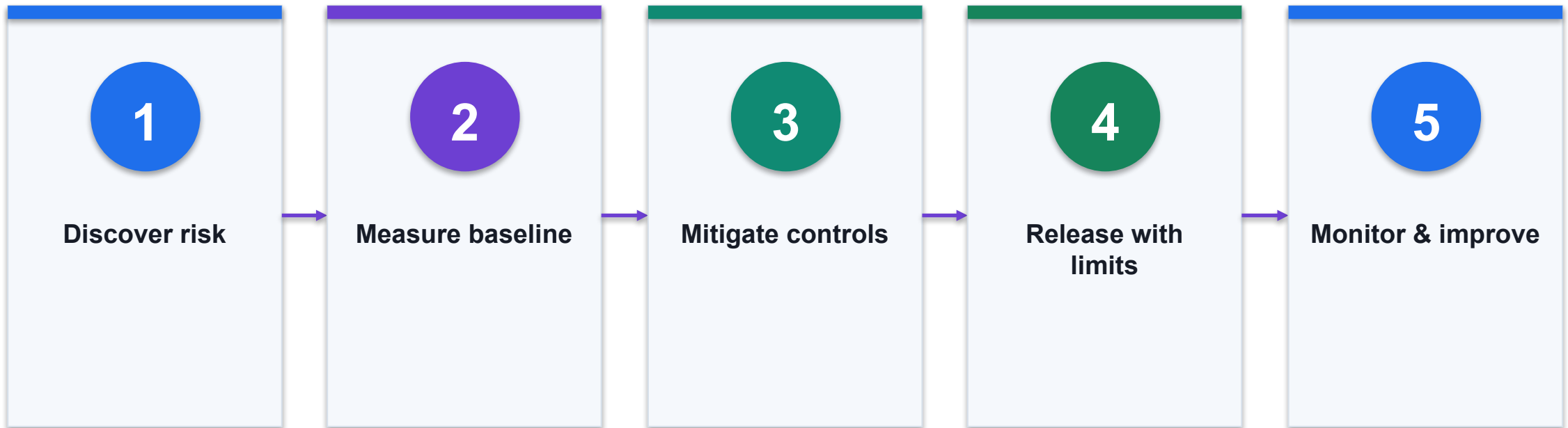
Responsible AI, content safety, and governance · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Responsible AI, content safety, and governance · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Responsible AI, content safety, and governance · exact contract

Contract	Exact value / field	Engineering consequence
Hate / Sexual / Violence / Self-harm	severity 0–6	Block at policy threshold
Prompt attack	direct / indirect	Reject, sanitize, or isolate tools
PII	category + offset + confidence	Redact or route to reviewer
Decision record	model, dataset, metric, owner	Versioned audit evidence

Responsible AI, content safety, and governance · implementation excerpt

```
POST /contentsafety/text:analyze?api-version=2024-09-01
{
  "text": user_text,
  "categories": ["Hate", "Violence", "SelfHarm",
"Sexual"],
  "outputType": "FourSeverityLevels"
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

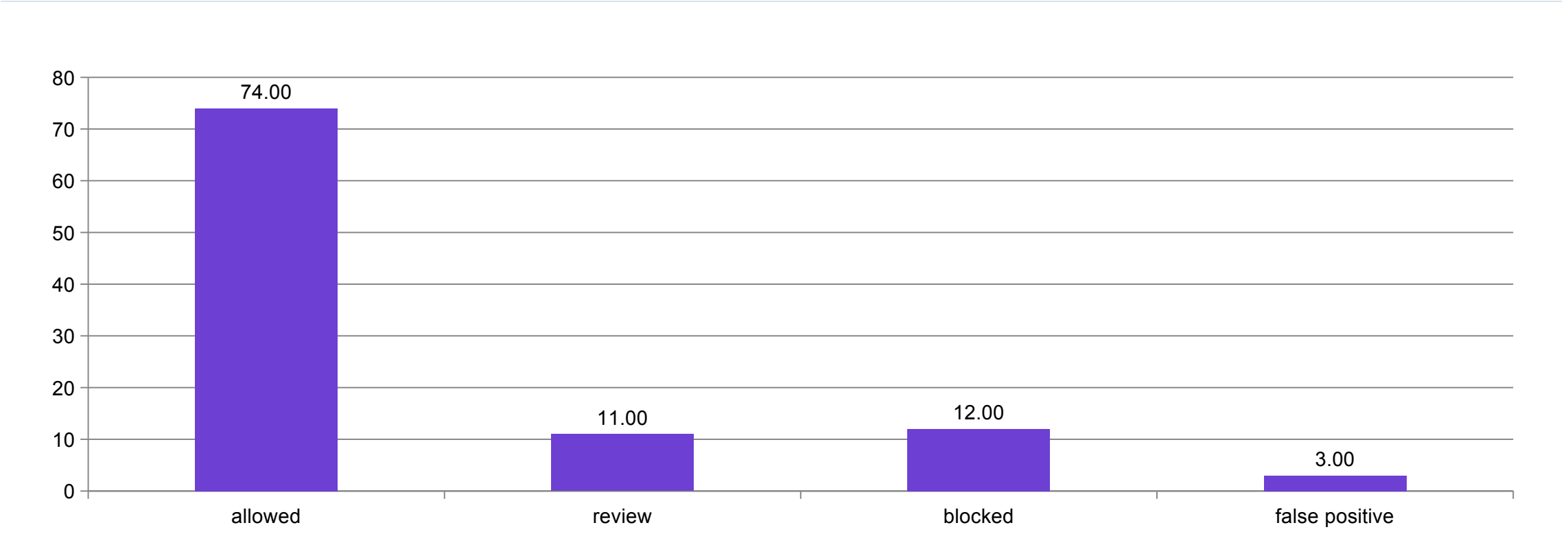
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

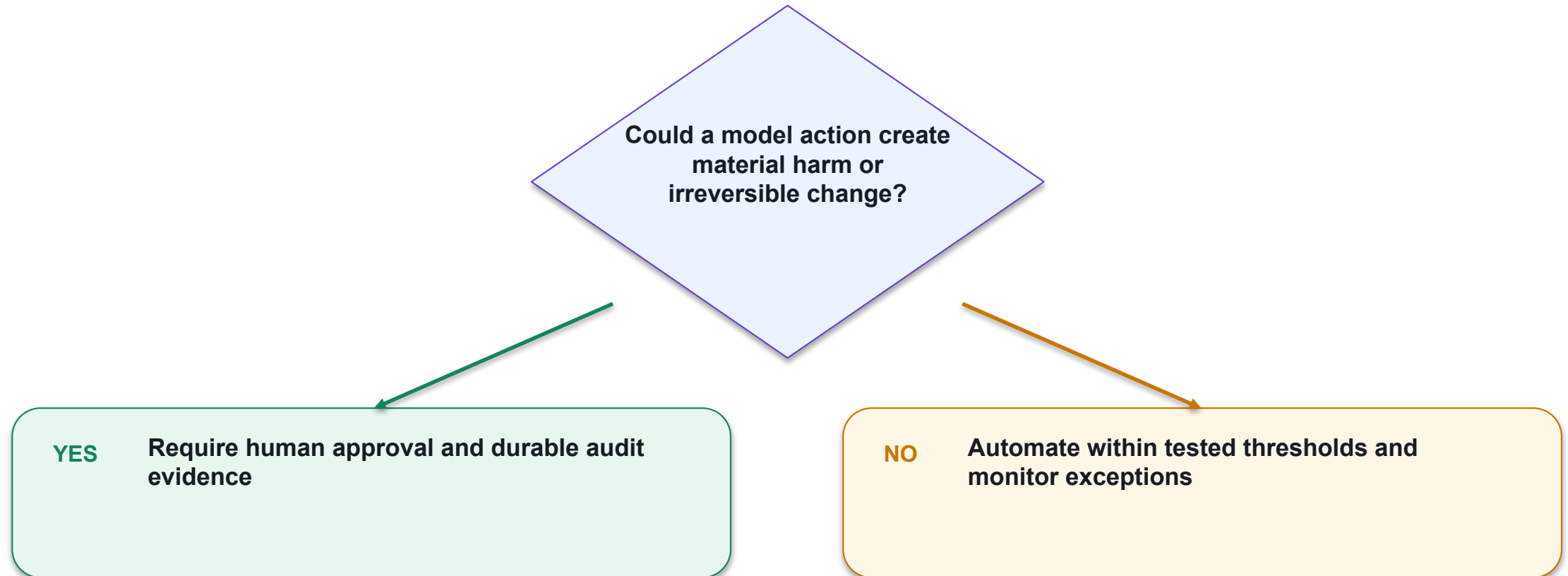
Handle auth, quota, timeout, and invalid-state paths.

Responsible AI, content safety, and governance · engineering metric



WHAT THE DATA SHOWS Indicative evaluation set: tune policy against both unsafe misses and legitimate-content false positives; one aggregate safety score is insufficient.

Responsible AI, content safety, and governance · decision rule



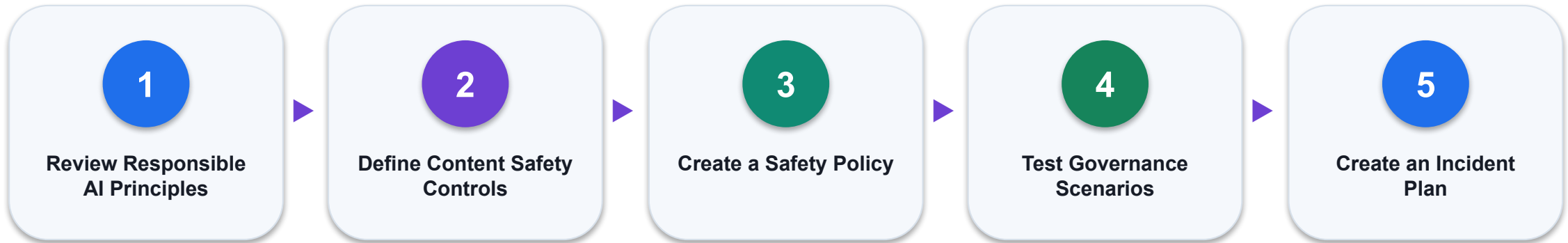
Responsible AI, content safety, and governance · failure and control

 Over-blocking LIKELY CAUSE Threshold too conservative CONTROL / RECOVERY Segment policy by channel and risk	 Jailbreak LIKELY CAUSE Instruction hierarchy compromised CONTROL / RECOVERY Isolate untrusted content and tools	 Data leak LIKELY CAUSE Prompt/output contains sensitive data CONTROL / RECOVERY Redact, minimize retention, restrict logging	 No owner LIKELY CAUSE Incident cannot be escalated CONTROL / RECOVERY Name response owner and SLA
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 02: Responsible AI, Content Safety, Governance

Mechanism focus: Responsible AI, content safety, and governance



YOU WILL PRODUCE You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

Complete click-by-click procedure: [Learner Guide · Lab 02 · labs/lab-02-responsible-ai-content-safety-governance/README.md](#)

Lab 02: acceptance evidence

PASS WHEN

Evidence pack: risk register, test dataset, per-category confusion matrix, filter configuration, exception workflow, and incident rehearsal result.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

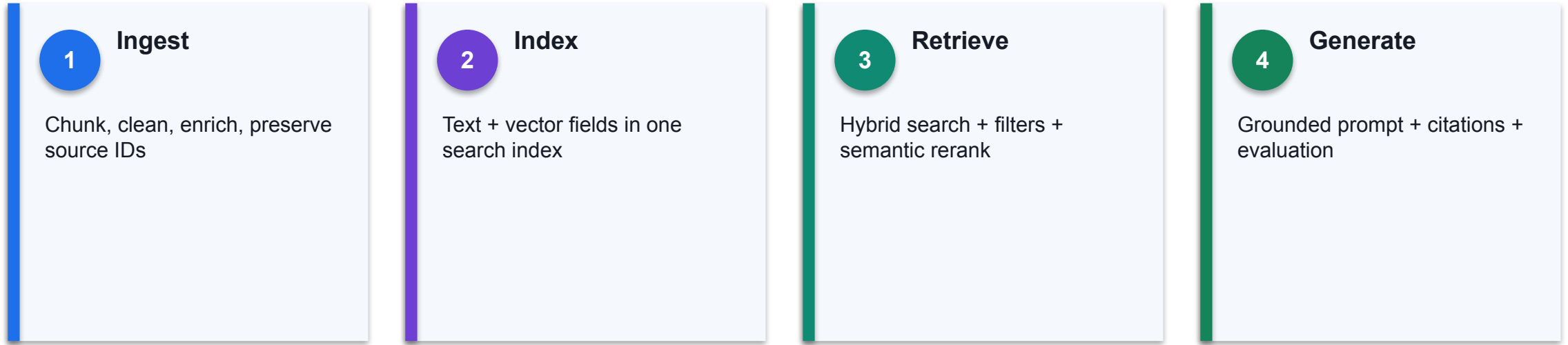
TOPIC 03

03

Generative AI apps and grounded RAG

GENERATIVE AI & AGENTS · AI-103 30–35%

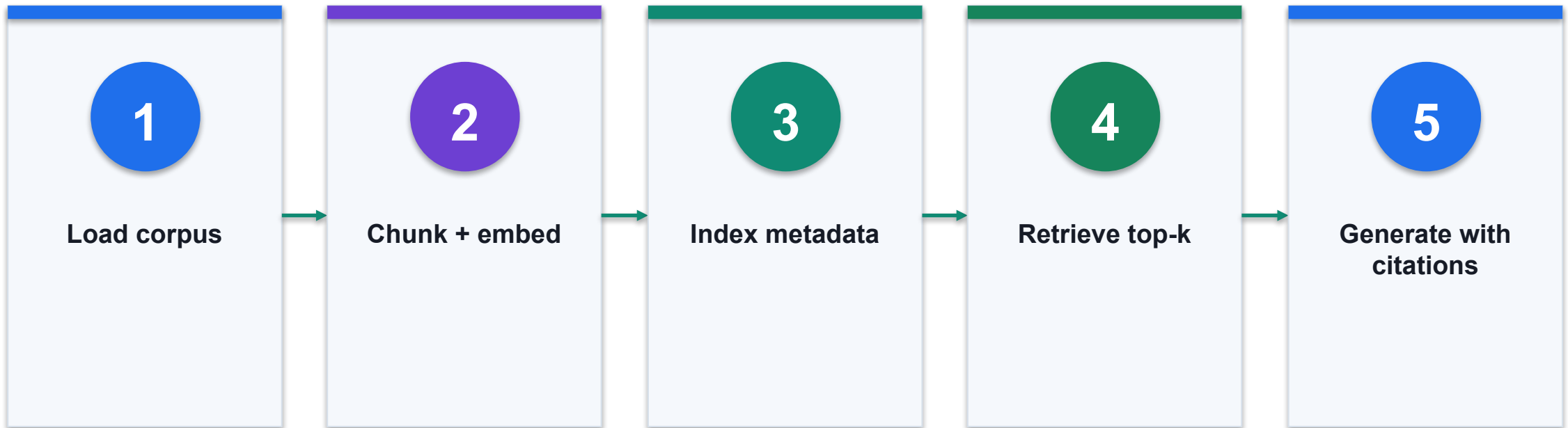
Generative AI apps and grounded RAG · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Generative AI apps and grounded RAG · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Generative AI apps and grounded RAG · exact contract

Contract	Exact value / field	Engineering consequence
chunk_id	string, key	Stable evidence identifier
content_vector	Collection(Edm.Single)	Dimensions match embedding model
filter	tenant_id eq 'T-42'	Security boundary before generation
top	5–20	Recall/latency trade-off

Generative AI apps and grounded RAG · implementation excerpt

```
response = openai.responses.create(
    model=DEPLOYMENT,
    input=[{"role": "user", "content": question}],
    tools=[{"type": "file_search",
            "vector_store_ids": [STORE_ID]}]
)
print(response.output_text)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

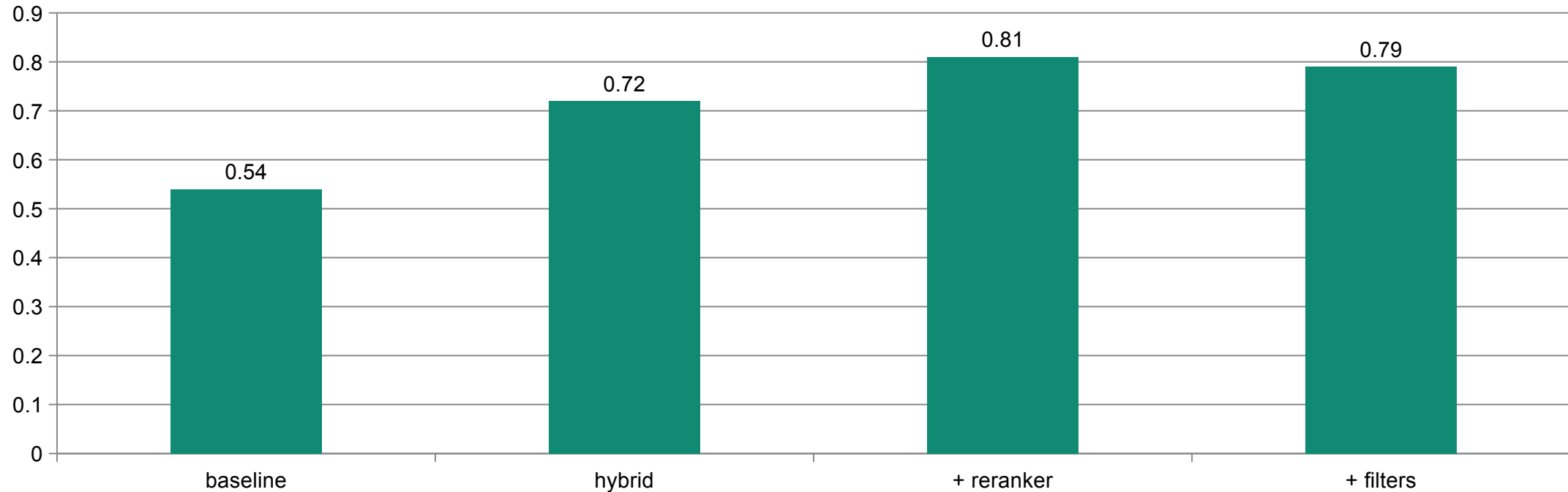
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

Handle auth, quota, timeout, and invalid-state paths.

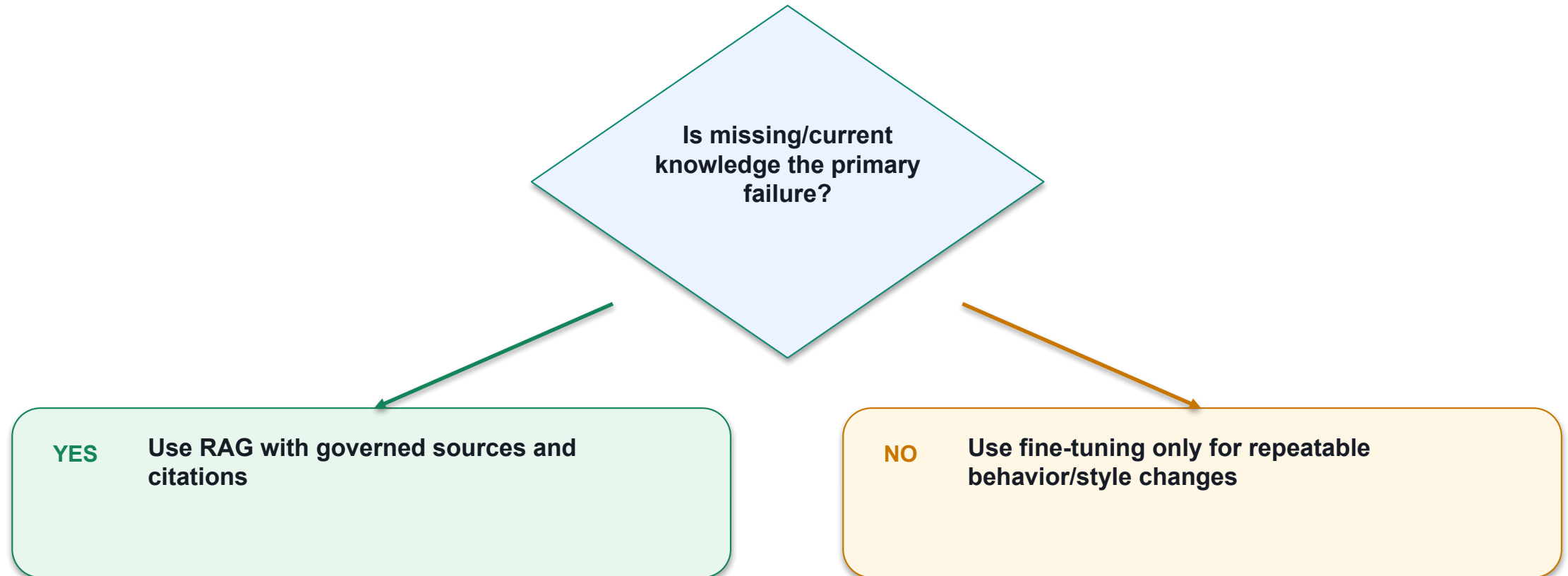
Generative AI apps and grounded RAG · engineering metric



WHAT THE DATA SHOWS

Indicative grounded-answer score: hybrid retrieval improves recall; reranking improves ordering; security filters may reduce recall and must be measured per tenant.

Generative AI apps and grounded RAG · decision rule



Generative AI apps and grounded RAG · failure and control

 Hallucination LIKELY CAUSE Context absent or ignored CONTROL / RECOVERY Require citations; answer 'not found'	 Dimension error LIKELY CAUSE Index and query embeddings differ CONTROL / RECOVERY Use one embedding model/config	 Leakage LIKELY CAUSE Filter applied after retrieval CONTROL / RECOVERY Filter at query/index boundary	 Stale corpus LIKELY CAUSE Indexer watermark failed CONTROL / RECOVERY Track source version and last-success time
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 03: Generative AI, Foundry, Azure OpenAI, Foundry Workflow, RAG

Mechanism focus: Generative AI apps and grounded RAG



YOU WILL PRODUCE

You should have model selection notes, prompt templates, tool-augmented workflow diagram, RAG design, and evaluation table.

Complete click-by-click procedure: [Learner Guide · Lab 03 · labs/lab-03-generative-ai-foundry-openai-prompt-flow-rag/README.md](#)

Lab 03: acceptance evidence

PASS WHEN

Evidence pack: chunk sample, index schema, hybrid query trace, cited answer, groundedness score, latency, and a negative test with no supporting evidence.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

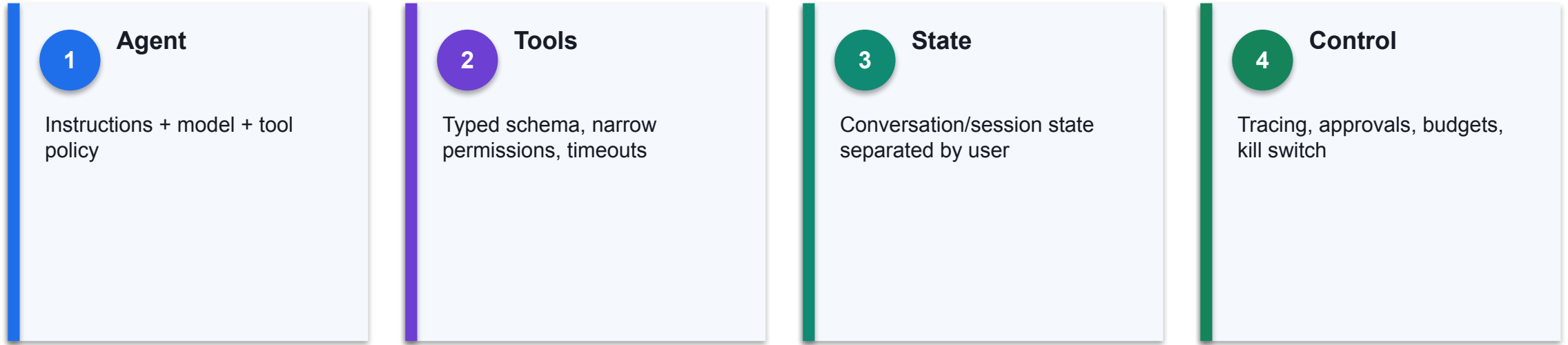
TOPIC 04

04

Agents, tools, memory, and multi-agent control

GENERATIVE AI & AGENTS · AI-103 30–35%

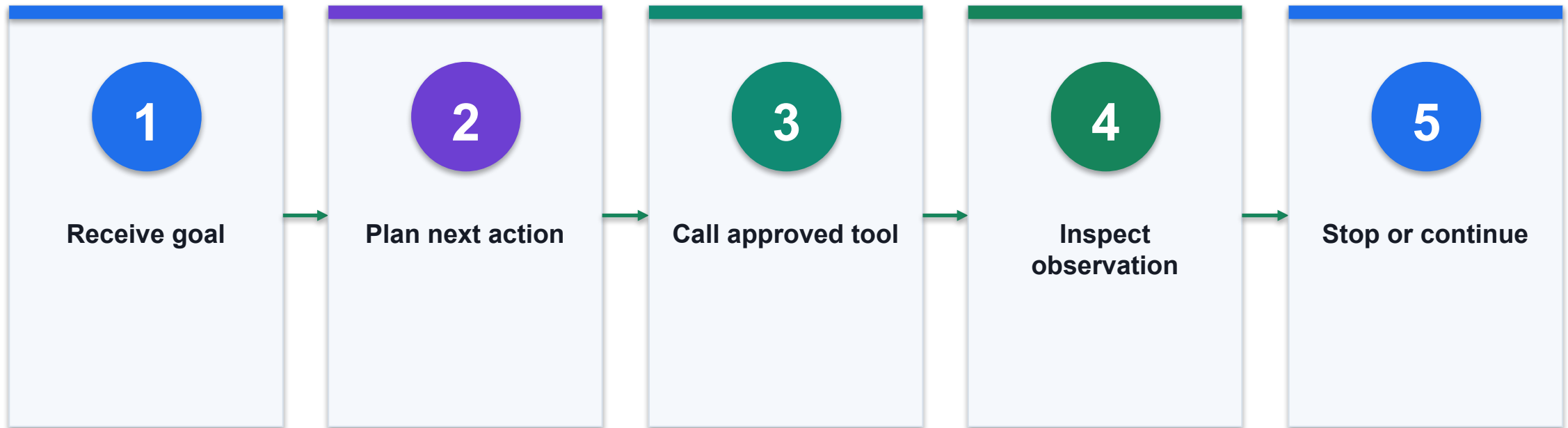
Agents, tools, memory, and multi-agent control · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Agents, tools, memory, and multi-agent control · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Agents, tools, memory, and multi-agent control · exact contract

Contract	Exact value / field	Engineering consequence
Tool schema	name + JSON parameters	Reject unknown arguments
Identity	Dedicated agent principal	No shared admin identity
Budget	max turns / tokens / cost	Hard termination condition
Trace	span, tool call, result, decision	Replayable evidence

Agents, tools, memory, and multi-agent control · implementation excerpt

```
agent = project.agents.create_version(
    agent_name="support-triage",
    definition=PromptAgentDefinition(
        model=MODEL_DEPLOYMENT,
        instructions=SYSTEM_POLICY,
        tools=[search_tool, ticket_tool]))
print(agent.id, agent.version)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

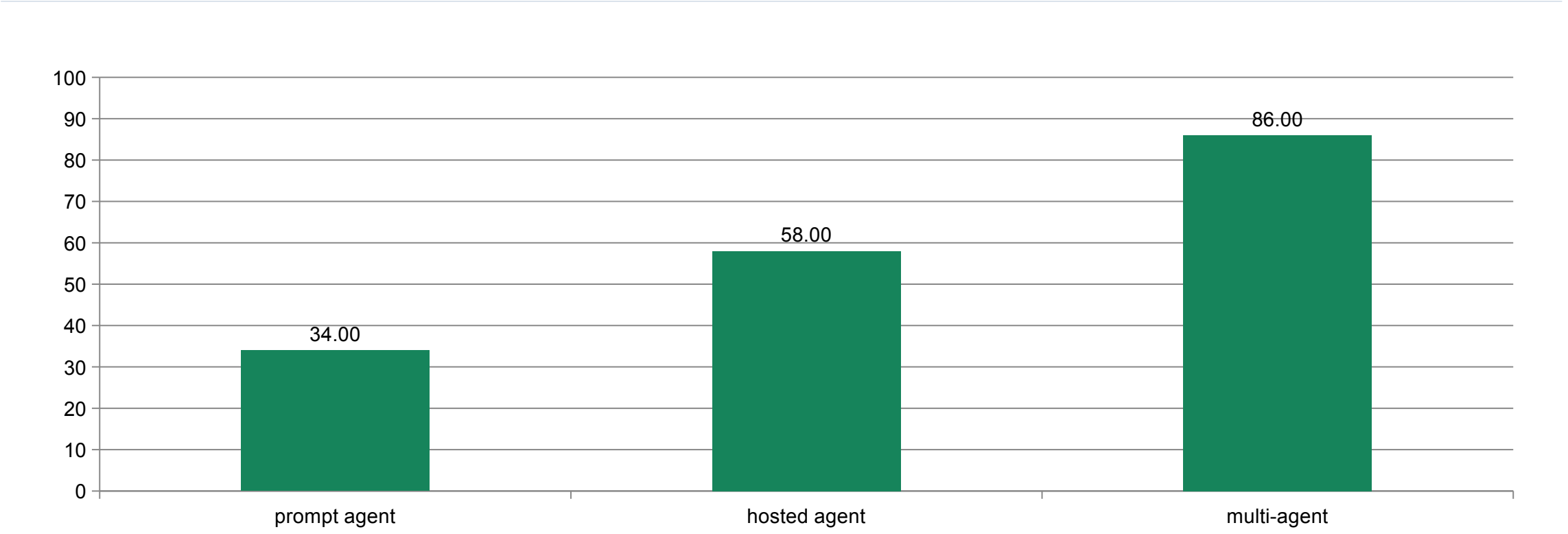
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

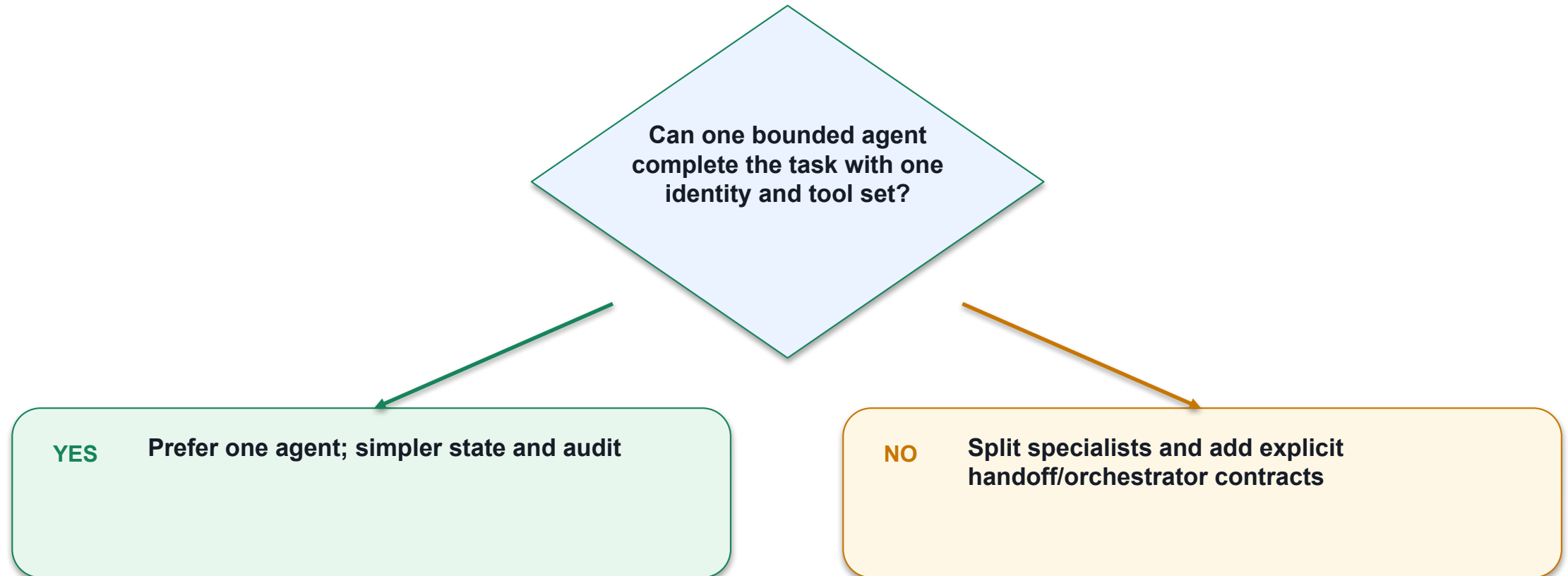
Handle auth, quota, timeout, and invalid-state paths.

Agents, tools, memory, and multi-agent control · engineering metric



WHAT THE DATA SHOWS Indicative operational complexity index: add hosting and orchestration only when autonomy, custom code, or specialization justifies extra identity, state, and observability controls.

Agents, tools, memory, and multi-agent control · decision rule



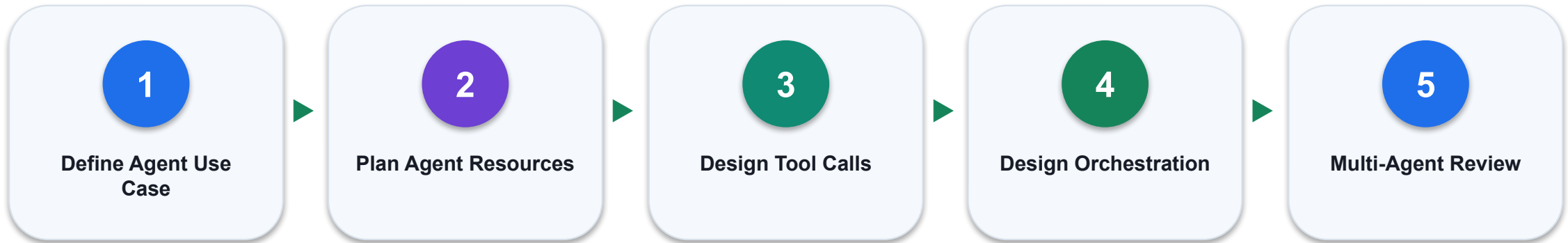
Agents, tools, memory, and multi-agent control · failure and control

 Tool abuse LIKELY CAUSE Broad permissions or weak schema CONTROL / RECOVERY Allow-list actions and validate arguments	 Loop LIKELY CAUSE No stop condition CONTROL / RECOVERY Turn/time/cost budgets	 Cross-user state LIKELY CAUSE Memory partition failure CONTROL / RECOVERY Key state by tenant + user + session	 Opaque handoff LIKELY CAUSE Agent loses context/owner CONTROL / RECOVERY Typed handoff payload + trace link
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 04: Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Mechanism focus: Agents, tools, memory, and multi-agent control



YOU WILL PRODUCE You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

Complete click-by-click procedure: [Learner Guide · Lab 04 · labs/lab-04-agentic-ai-foundry-agent-service/README.md](#)

Lab 04: acceptance evidence

PASS WHEN

Evidence pack: agent definition/version, tool schemas, least-privilege identity, approval gate, trace showing tool call/result, budget exhaustion test, and handoff test.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

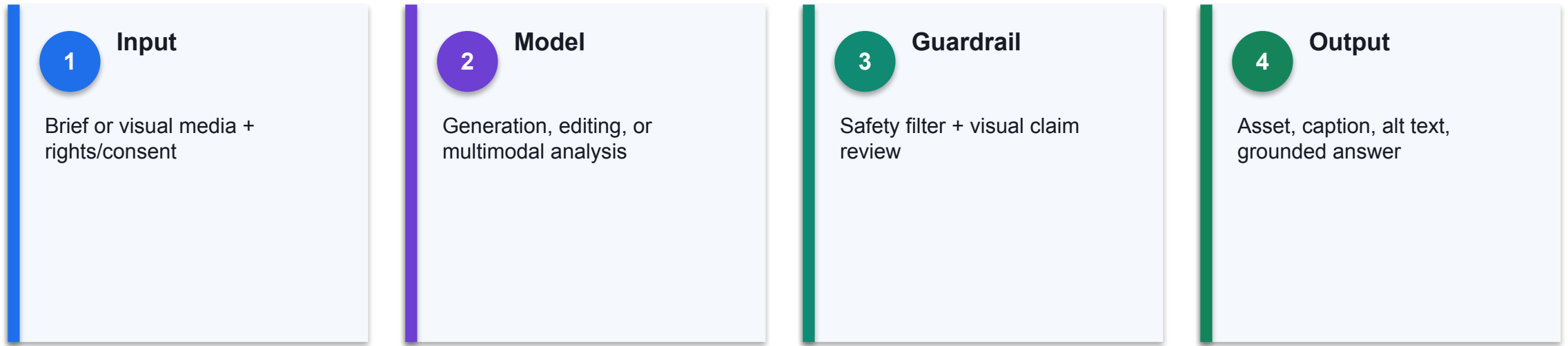
TOPIC 05

05

Image/video generation and multimodal understanding

COMPUTER VISION · AI-103 10–15%

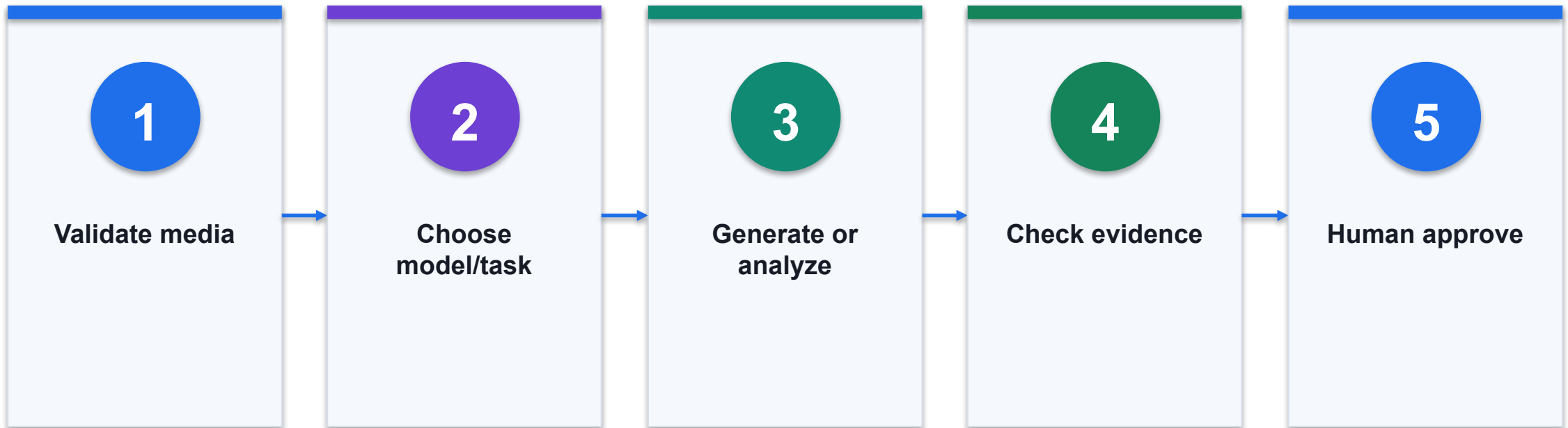
Image/video generation and multimodal understanding · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Image/video generation and multimodal understanding · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Image/video generation and multimodal understanding · exact contract

Contract	Exact value / field	Engineering consequence
asset_id	stable source/output ID	Provenance and rollback
model_version	pinned deployment/version	Reproducible evaluation
claim	text + source evidence	Reject unsupported product facts
review	approved + reason + owner	Publication gate

Image/video generation and multimodal understanding · implementation excerpt

```
brief = {"capacity_ml": 500, "allowed_claims":
["reusable"]}
claims = ["reusable", "48-hour cooling"]
unsupported = set(claims) -
set(brief["allowed_claims"])
assert not unsupported, f"Reject unsupported claims:
{unsupported}"
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

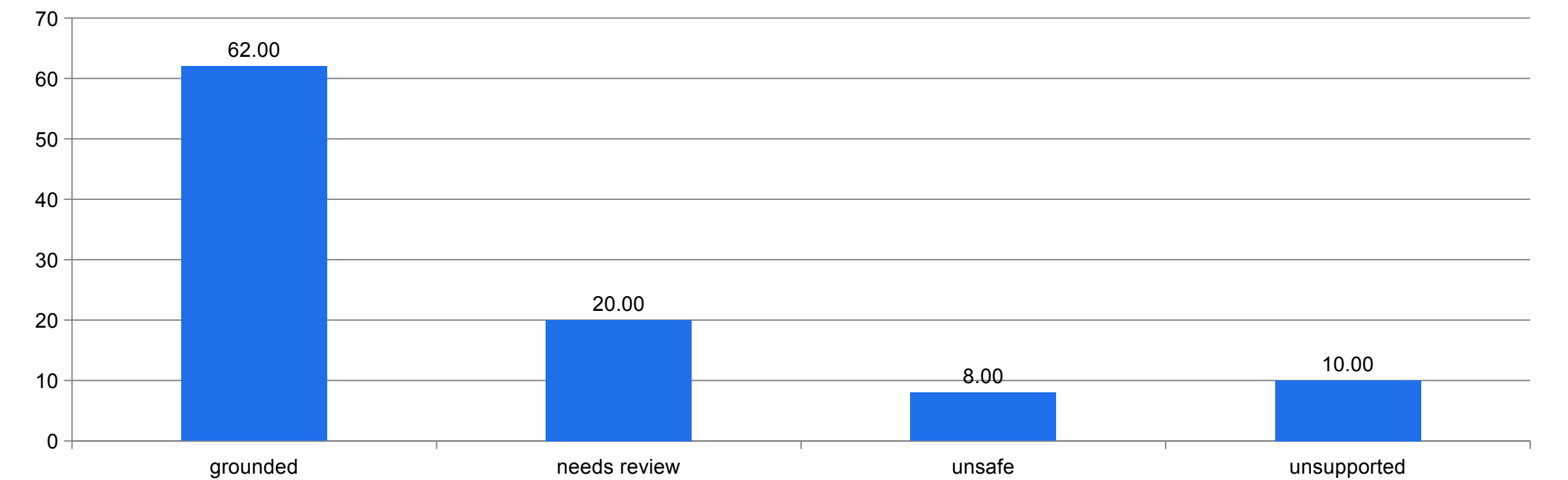
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

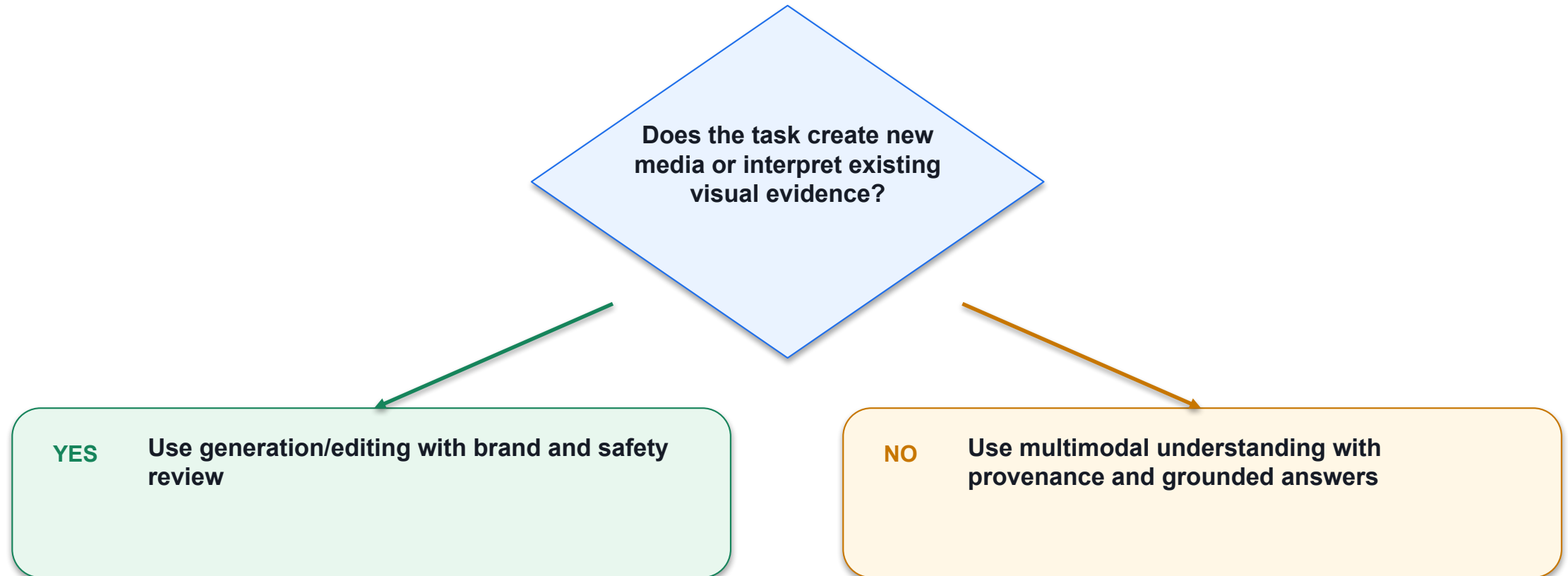
Handle auth, quota, timeout, and invalid-state paths.

Image/video generation and multimodal understanding · engineering metric



WHAT THE DATA SHOWS Illustrative review set (%): generation quality alone does not grant publication; groundedness, safety, accessibility, and human approval are separate gates.

Image/video generation and multimodal understanding · decision rule



Image/video generation and multimodal understanding - failure and control

 Fabricated claim LIKELY CAUSE Output exceeds source brief CONTROL / RECOVERY Compare with approved attributes	 Prompt injection LIKELY CAUSE Text embedded in image instructs agent CONTROL / RECOVERY Treat media as untrusted data	 Accessibility LIKELY CAUSE Alt text omits visual facts CONTROL / RECOVERY Evaluate against source image	 Rights LIKELY CAUSE Reference media lacks permission CONTROL / RECOVERY Validate provenance and consent
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 05: Image and Video Generation, Multimodal Understanding

Mechanism focus: Image/video generation and multimodal understanding



YOU WILL PRODUCE

A complete submission has the decision table, prompt and model record, `review.json`, passing local validation, architecture, and one rejected output with a cited reason. Live Foundry evidence is optional and must be labeled separately.

Complete click-by-click procedure: [Learner Guide · Lab 05 · labs/lab-05-image-video-generation-multimodal/README.md](#)

Lab 05: acceptance evidence

PASS WHEN

Evidence pack: task selection, source brief, model/version, prompt, generated or sample output, caption and alt text, claim/safety review, and human publication decision.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

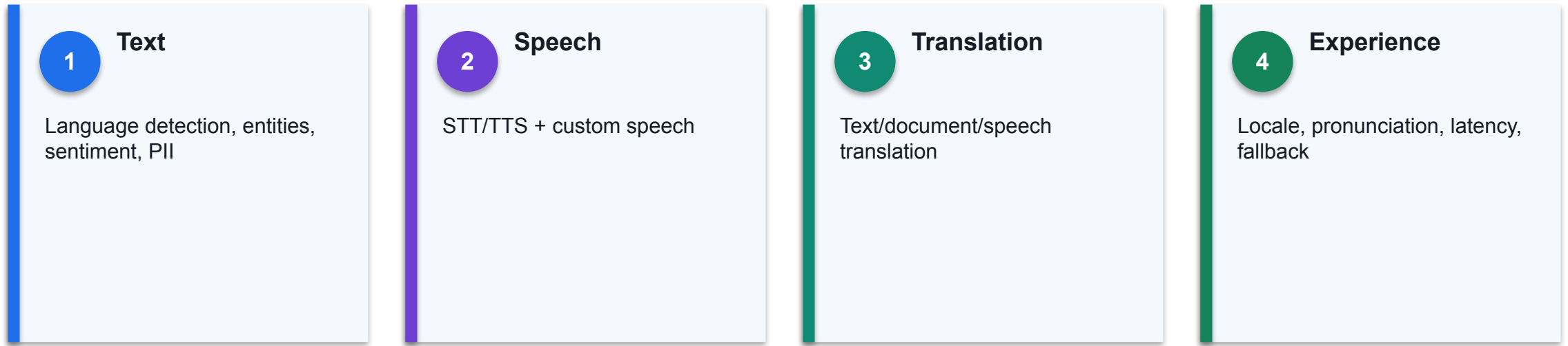
TOPIC 06

06

Language, speech, translation, and SSML

TEXT ANALYSIS · AI-103 10–15%

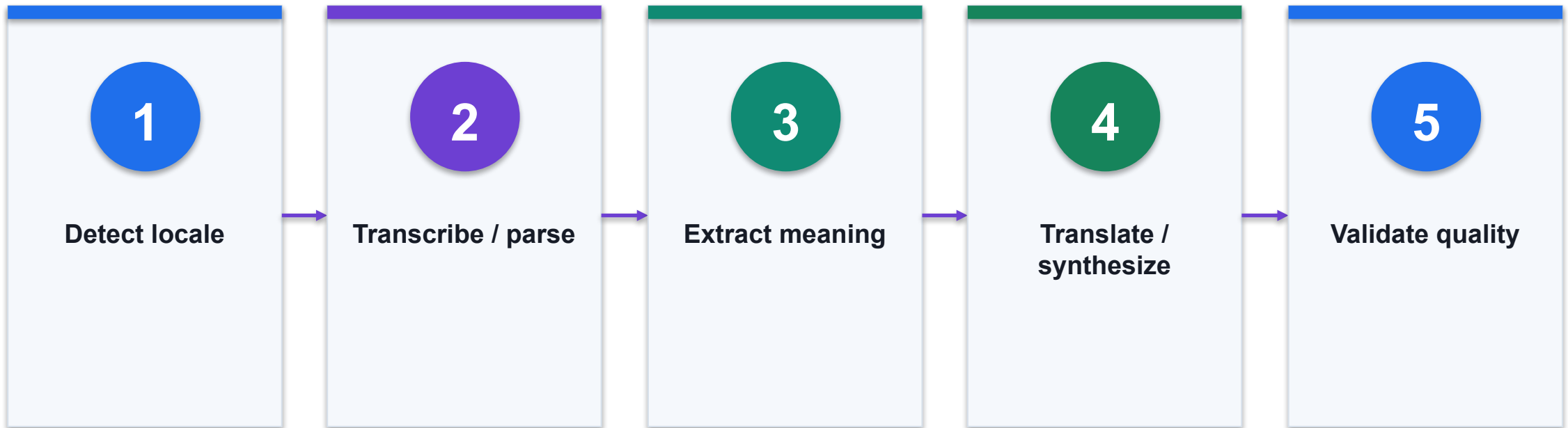
Language, speech, translation, and SSML · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Language, speech, translation, and SSML · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Language, speech, translation, and SSML · exact contract

Contract	Exact value / field	Engineering consequence
language	en-SG / zh-Hans / auto	Affects model and voice
offset + length	Unicode text span	Preserve source mapping
SSML	voice, prosody, phoneme, break	Controlled speech output
recognition result	text + reason + duration	Handle NoMatch/Canceled

Language, speech, translation, and SSML · implementation excerpt

```
<speak version="1.0" xml:lang="en-SG">
  <voice name="en-SG-LunaNeural">
    Your request is <prosody
rate="-5%">approved</prosody>.
  </voice>
</speak>
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

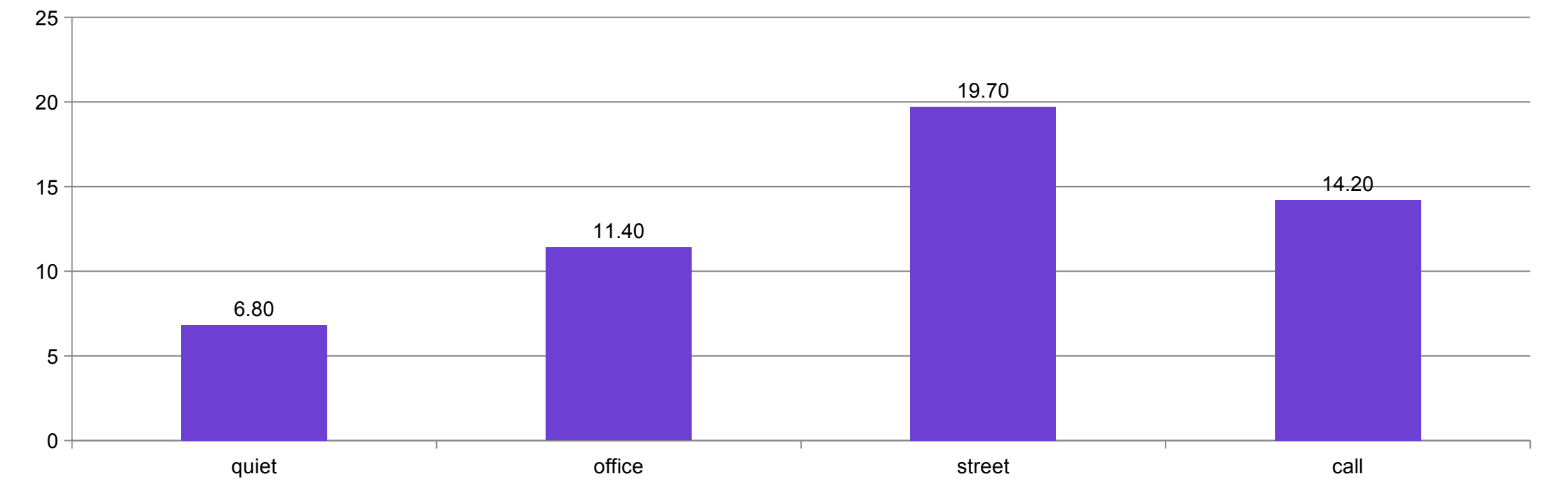
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

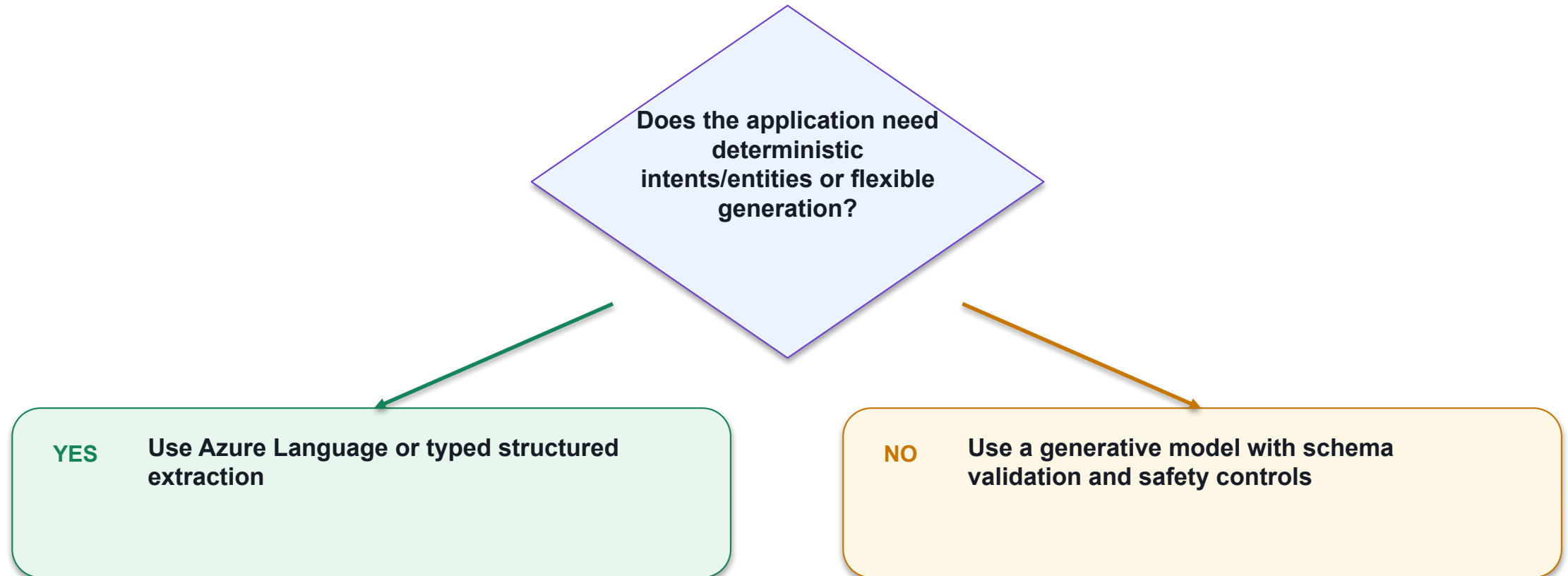
Handle auth, quota, timeout, and invalid-state paths.

Language, speech, translation, and SSML · engineering metric



WHAT THE DATA SHOWS Indicative word error rate (%): acoustic context changes recognition quality. Test the real channel and vocabulary; do not accept a studio-only benchmark.

Language, speech, translation, and SSML · decision rule



Language, speech, translation, and SSML - failure and control

 NoMatch LIKELY CAUSE Audio/language mismatch CONTROL / RECOVERY Inspect reason and locale	 PII leak LIKELY CAUSE Transcript retained CONTROL / RECOVERY Redact before logging	 Bad pronunciation LIKELY CAUSE Voice/phoneme mismatch CONTROL / RECOVERY Use SSML phoneme/lexicon	 Translation drift LIKELY CAUSE Domain terms mistranslated CONTROL / RECOVERY Glossary/custom model + review
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 06: NLP, Language, Speech, Translation

Mechanism focus: Language, speech, translation, and SSML



YOU WILL PRODUCE You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.

Complete click-by-click procedure: [Learner Guide · Lab 06 · labs/lab-06-nlp-language-speech-translation/README.md](#)

Lab 06: acceptance evidence

PASS WHEN

Evidence pack: multilingual samples, PII span/redaction, recognition reasons, SSML output, word-error-rate sample, and human review for domain terms.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 00

00

Day 2

Custom language · search · information extraction · capstone

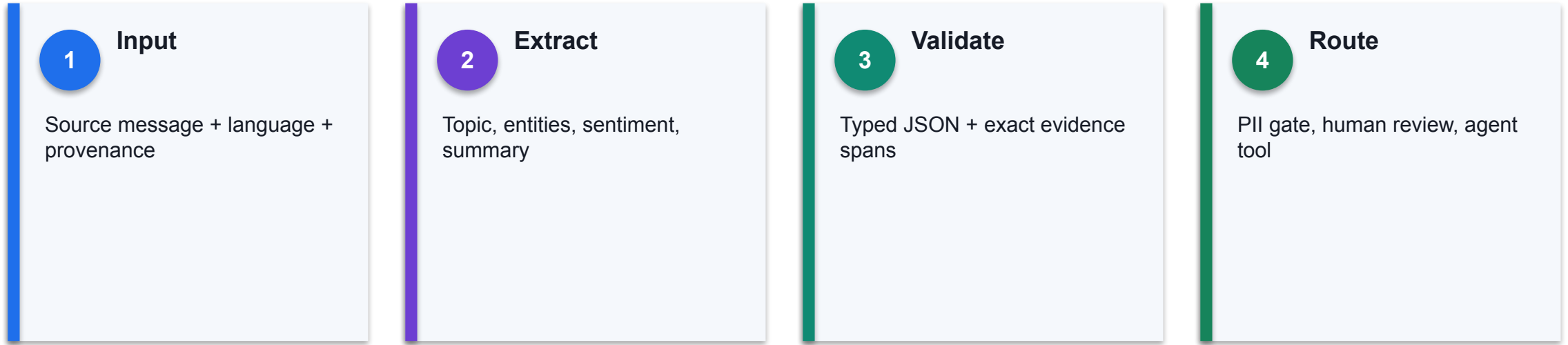
TOPIC 07

07

Generative text analysis and translation

TEXT ANALYSIS · AI-103 10–15%

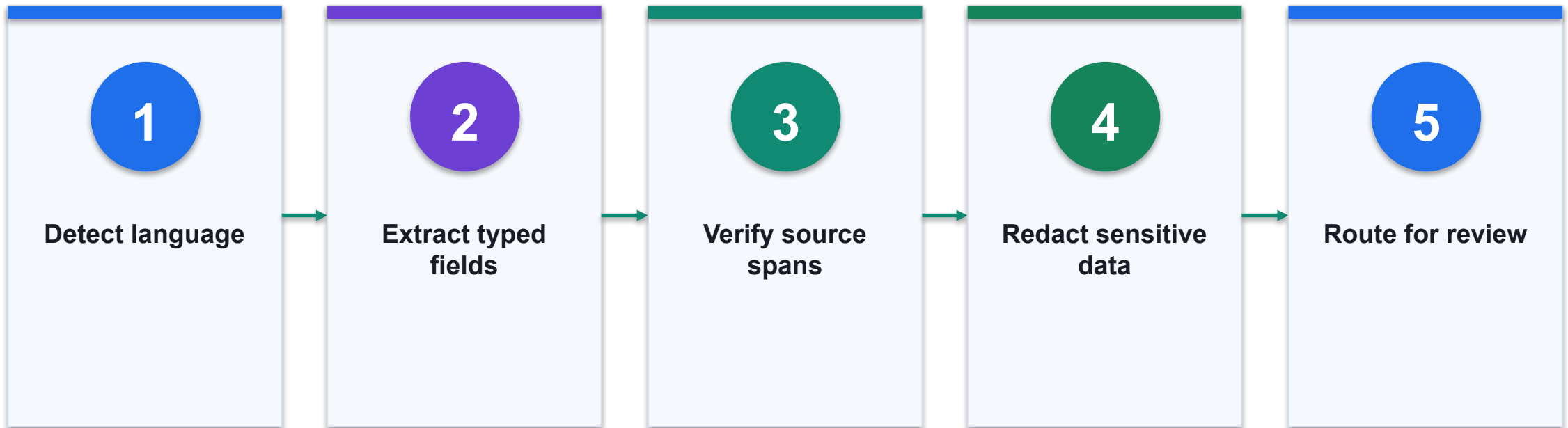
Generative text analysis and translation · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Generative text analysis and translation · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Generative text analysis and translation · exact contract

Contract	Exact value / field	Engineering consequence
message_id	stable source ID	Traceability
evidence_span	exact source substring	Reject unsupported claims
sensitive_content	boolean	Review and redact
review_required	boolean	No unsafe auto-action

Generative text analysis and translation · implementation excerpt

```
record = {"message_id": "M1", "evidence_span": "refund  
for order 12345"}  
source = "I need a refund for order 12345."  
assert record["evidence_span"] in source  
# Validate every generated field before an agent uses  
it
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

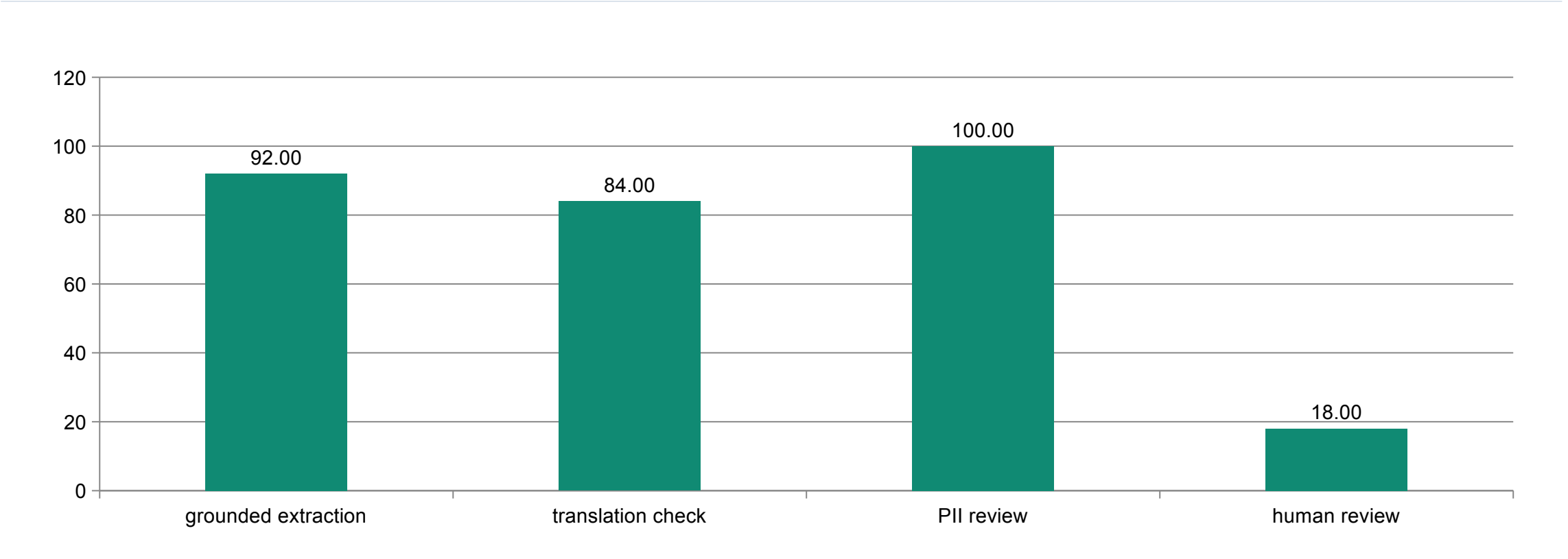
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

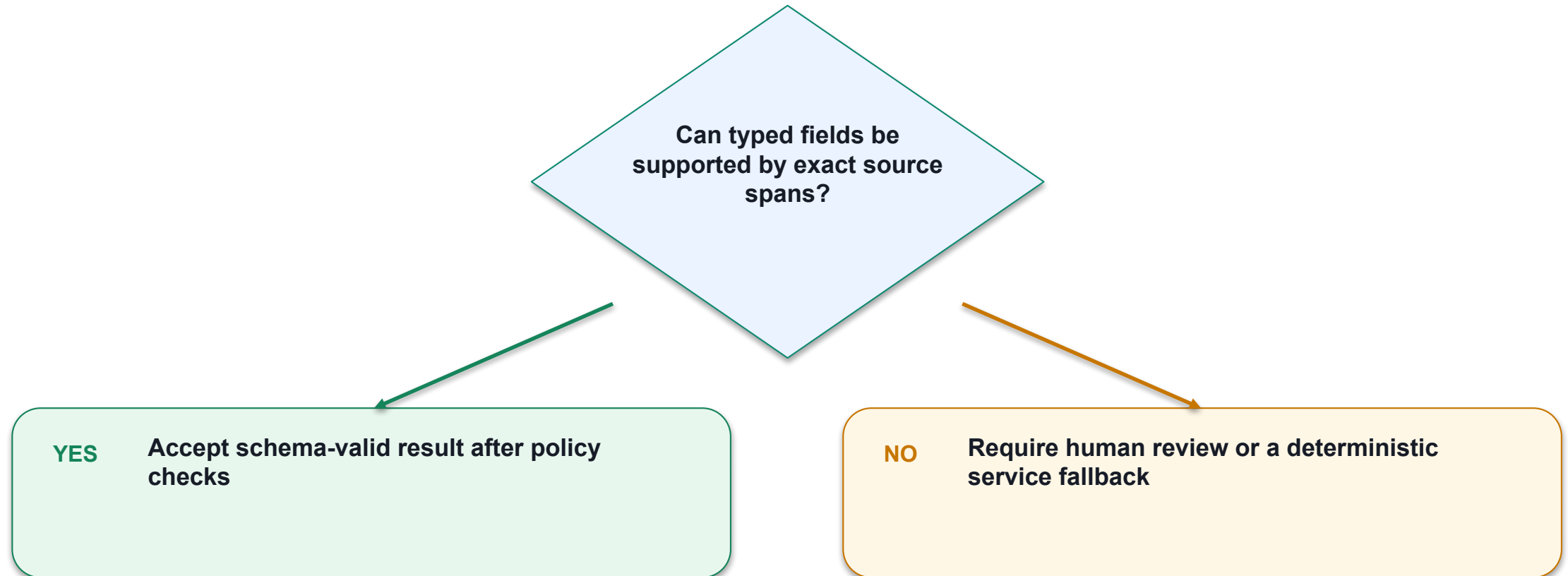
Handle auth, quota, timeout, and invalid-state paths.

Generative text analysis and translation · engineering metric



WHAT THE DATA SHOWS Illustrative evaluation set (%): report source-grounded extraction, translation fidelity, and sensitive-content handling separately.

Generative text analysis and translation · decision rule



Generative text analysis and translation · failure and control

 Invented entity LIKELY CAUSE Model guesses a customer fact CONTROL / RECOVERY Require exact evidence span	 PII leak LIKELY CAUSE Identifier enters logs or tool CONTROL / RECOVERY Redact before routing	 Translation drift LIKELY CAUSE Domain term changes meaning CONTROL / RECOVERY Compare source and translation	 Bad JSON LIKELY CAUSE Agent tool receives malformed output CONTROL / RECOVERY Validate types and required fields
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A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 07: Generative Text Analysis and Translation

Mechanism focus: Generative text analysis and translation



YOU WILL PRODUCE

Submit ``analysis.json``, the validator PASS output, prompt, translation comparison, and agent data-flow. Each record must have a source-bound evidence span, correct sensitive-content decision, and no invented entity. Live service evidence is optional and must be labeled separately.

Complete click-by-click procedure: [Learner Guide · Lab 07 · labs/lab-07-generative-text-analysis/README.md](#)

Lab 07: acceptance evidence

PASS WHEN

Evidence pack: synthetic source messages, typed JSON, exact spans, sensitive-content decision, translation comparison, validator output, and agent routing rule.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

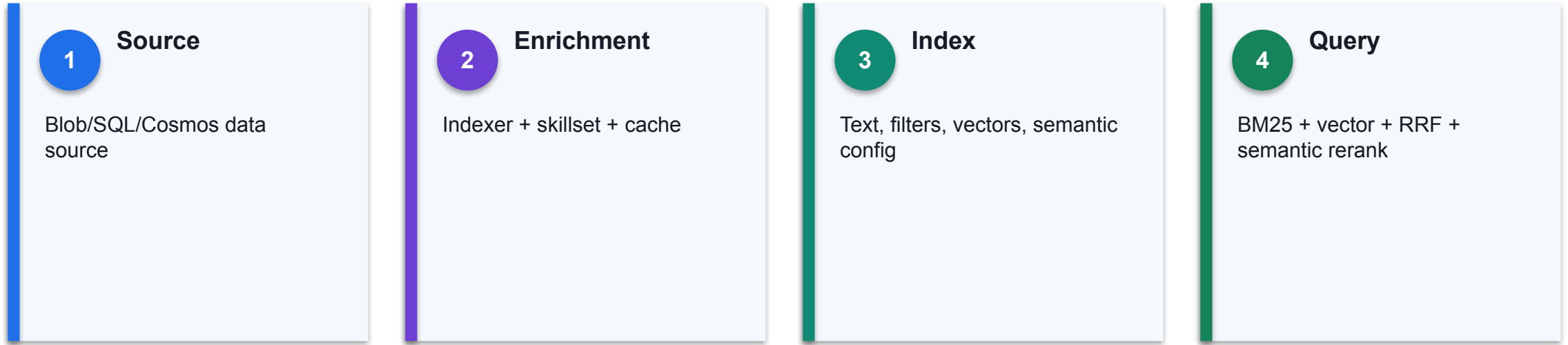
TOPIC 08

08

Knowledge retrieval, indexing, and grounding

INFORMATION EXTRACTION · AI-103 10–15%

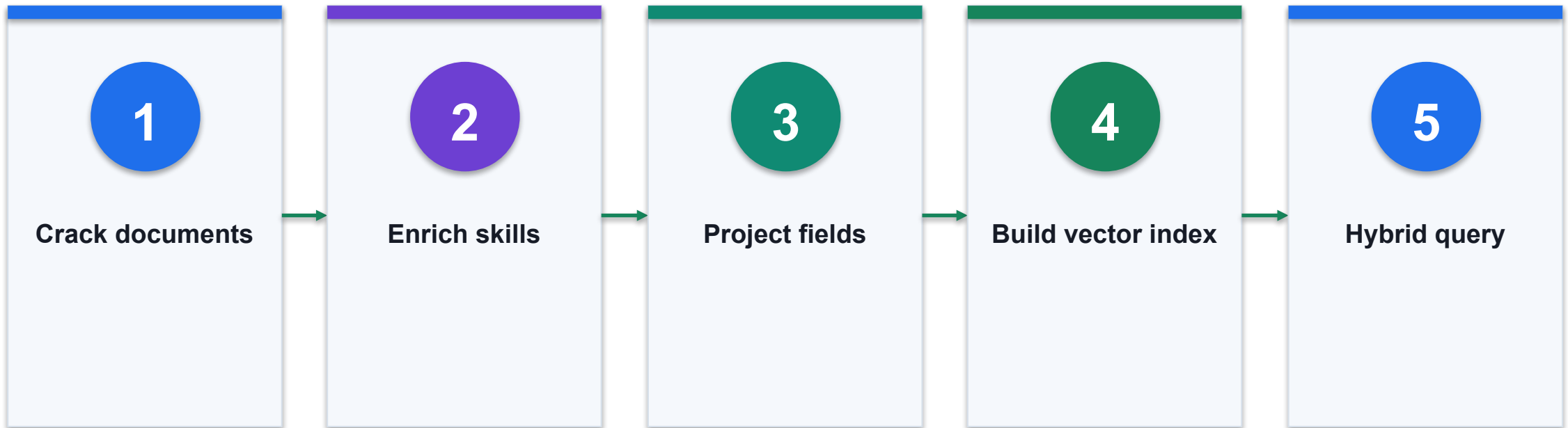
Knowledge retrieval, indexing, and grounding · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Knowledge retrieval, indexing, and grounding · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Knowledge retrieval, indexing, and grounding · exact contract

Contract	Exact value / field	Engineering consequence
key	Edm.String, key=true	Unique document/chunk ID
vector	Collection(Edm.Single)	Dimensions + HNSW profile
search/filter/select	query contract	Reduce exposure and payload
scores	@search.score / rerankerScore	Different scales; do not compare directly

Knowledge retrieval, indexing, and grounding · implementation excerpt

```
{
  "search": "maintenance policy",
  "vectorQueries": [{"kind": "text", "text":
"maintenance policy",
+      "fields": "content_vector"}],
  "queryType": "semantic",
  "filter": "tenant_id eq 'T-42'",
  "top": 5
}
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

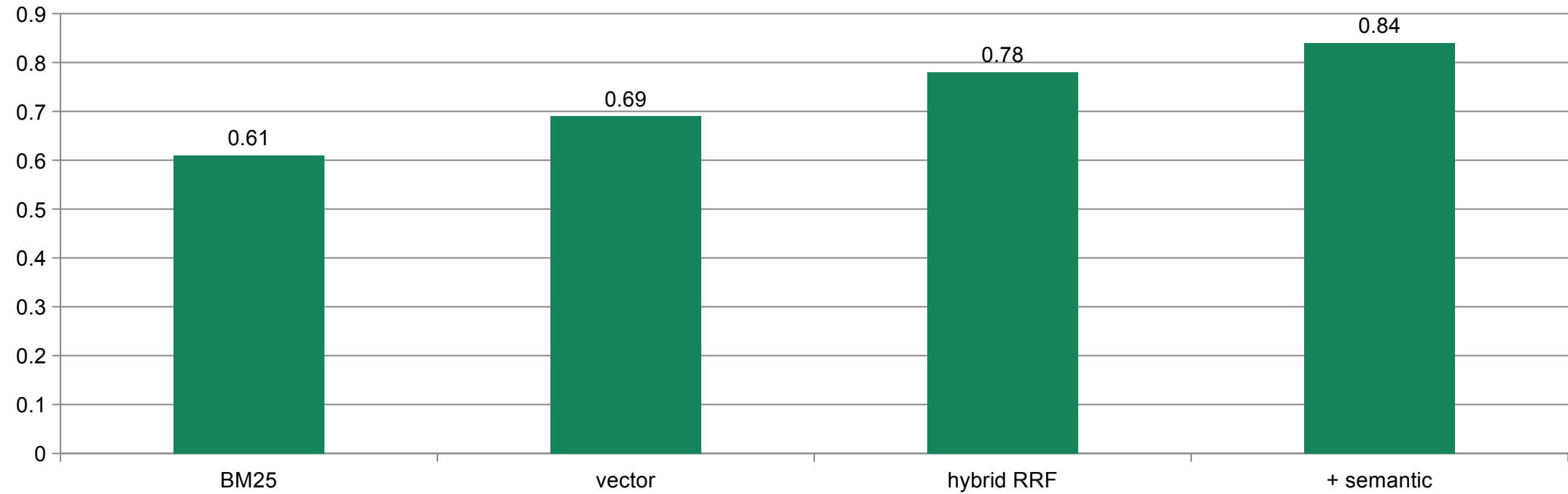
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

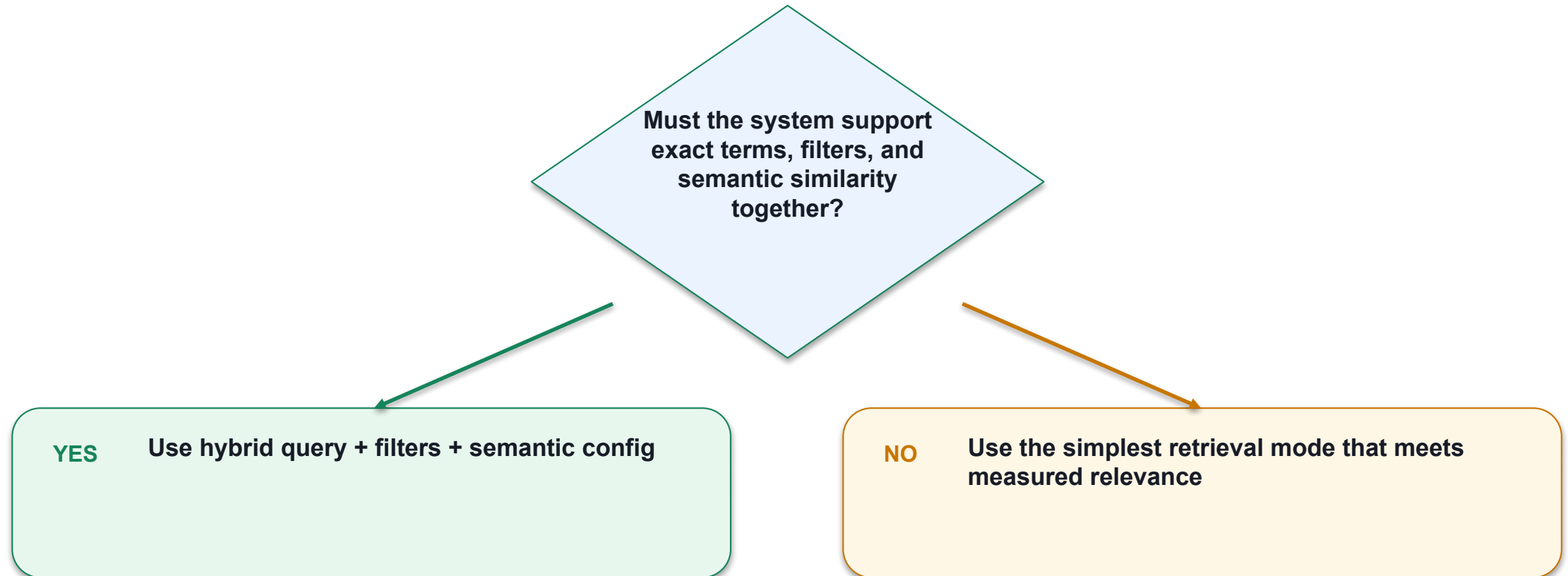
Handle auth, quota, timeout, and invalid-state paths.

Knowledge retrieval, indexing, and grounding · engineering metric



WHAT THE DATA SHOWS Indicative nDCG@5: hybrid retrieval often improves coverage, while semantic reranking improves ordering. Validate with your own judged query set.

Knowledge retrieval, indexing, and grounding · decision rule



Knowledge retrieval, indexing, and grounding · failure and control

 Indexer error LIKELY CAUSE Skill input path missing CONTROL / RECOVERY Inspect enrichment tree and field mappings	 Empty vector LIKELY CAUSE Embedding skill/vectorizer failed CONTROL / RECOVERY Check endpoint, dimensions, projection	 Poor relevance LIKELY CAUSE No judged query set CONTROL / RECOVERY Create labels and compare retrieval modes	 Data leak LIKELY CAUSE filterable security field absent CONTROL / RECOVERY Enforce tenant/ACL filter before results
--	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 08: Azure AI Search, Knowledge Mining, Vector Search

Mechanism focus: Knowledge retrieval, indexing, and grounding



YOU WILL PRODUCE

You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.

Complete click-by-click procedure: [Learner Guide · Lab 08 · labs/lab-08-ai-search-knowledge-mining-vector-search/README.md](#)

Lab 08: acceptance evidence

PASS WHEN

Evidence pack: index schema, data source/indexer/skillset JSON, successful run history, judged queries, hybrid trace, security-filter negative test, and knowledge-store projection.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 09

09

Document Intelligence and Content Understanding

INFORMATION EXTRACTION · AI-103 10–15%

Document Intelligence and Content Understanding



Input

Document/image/audio/video + content type

Analyzer

Prebuilt/custom/composed/Content Understanding

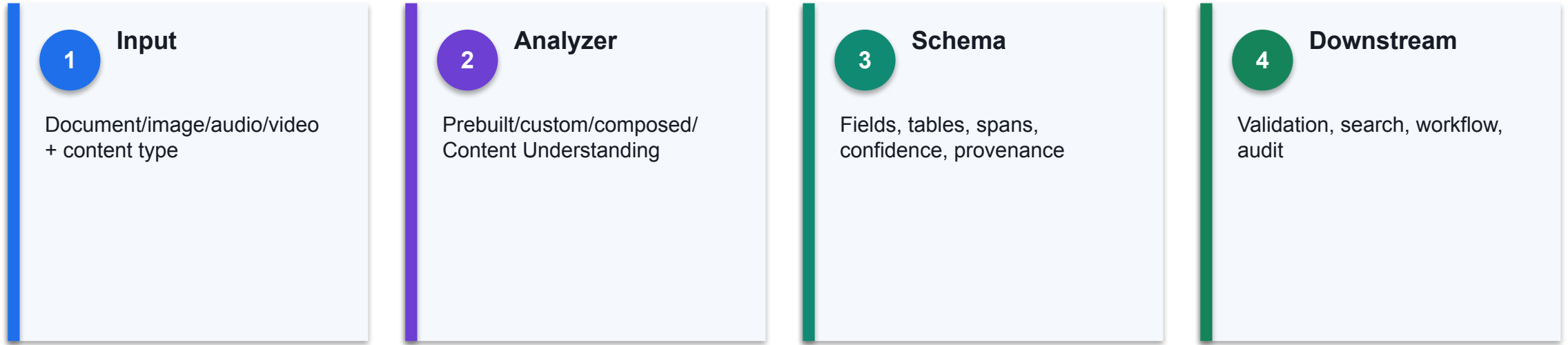
Schema

Fields, tables, spans, confidence, provenance

Downstream

Validation, search, workflow, audit

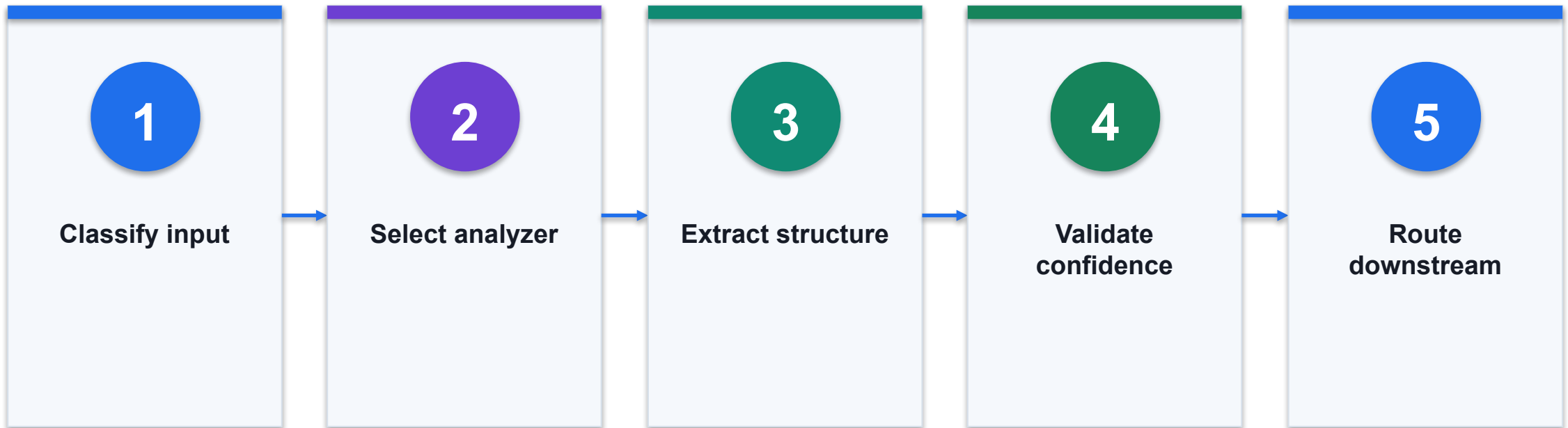
Document Intelligence and Content Understanding · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Document Intelligence and Content Understanding · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Document Intelligence and Content Understanding · exact contract

Contract	Exact value / field	Engineering consequence
modelId / analyzerId	prebuilt or versioned custom	Release identity
pages / spans	offset + length + polygon	Trace extracted value to source
confidence	field-level 0–1	Drive review threshold
operation-location	async result URL	Poll until succeeded/failed

Document Intelligence and Content Understanding - implementation excerpt

```
poller = client.begin_analyze_document(
    model_id="prebuilt-invoice",
    body=document_bytes,
    content_type="application/pdf")
result = poller.result()
for doc in result.documents:
    print(doc.fields["InvoiceTotal"].value_currency)
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

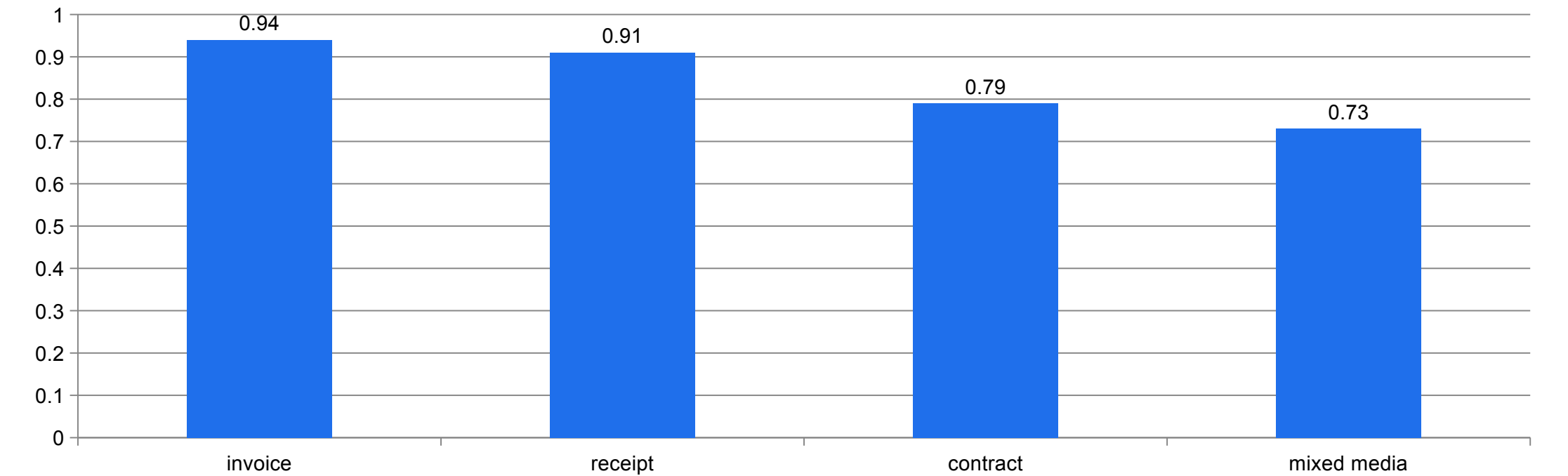
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

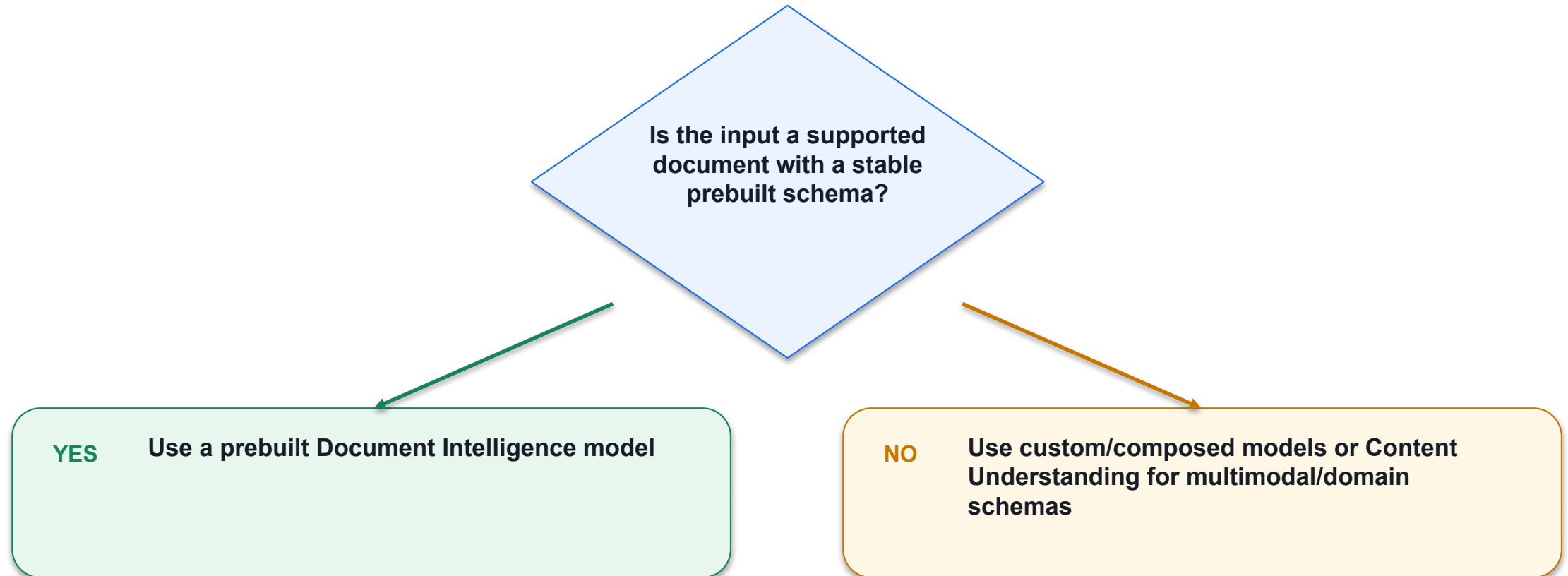
Handle auth, quota, timeout, and invalid-state paths.

Document Intelligence and Content Understanding - engineering metric



WHAT THE DATA SHOWS Indicative field F1: prebuilt models excel on supported formats; custom/composed models or Content Understanding address domain-specific and multimodal schemas.

Document Intelligence and Content Understanding - decision rule



Document Intelligence and Content Understanding - failure and control

 Wrong model LIKELY CAUSE Document type misclassified CONTROL / RECOVERY Classify first or use composed model	 Low confidence LIKELY CAUSE Layout/domain drift CONTROL / RECOVERY Human review + retraining sample	 Async timeout LIKELY CAUSE Polling/size/page limits CONTROL / RECOVERY Honor operation status and limits	 Lost provenance LIKELY CAUSE Value stored without span/page CONTROL / RECOVERY Persist source URI + region/span
---	--	---	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 09: Document Intelligence and Content Understanding

Mechanism focus: Document Intelligence and Content Understanding



YOU WILL PRODUCE

You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.

Complete click-by-click procedure: [Learner Guide · Lab 09 · labs/lab-09-document-intelligence-content-understanding/README.md](#)

Lab 09: acceptance evidence

PASS WHEN

Evidence pack: analyzer/model ID, sample result JSON, source polygons/spans, field-level metrics, review threshold, failure sample, and downstream search/workflow record.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 10

10

Capstone: production AI solution decision record

CAPSTONE · INTEGRATE, OPERATE, AND IMPROVE

Capstone: production AI solution decision record



Experience

API/UI + accessibility + fallback

Intelligence

Model/agent + retrieval + extraction

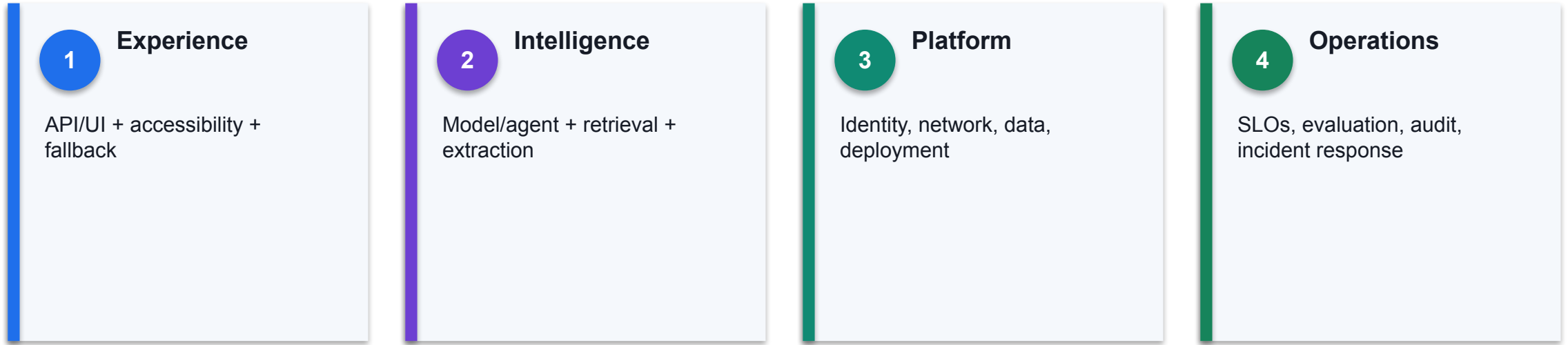
Platform

Identity, network, data, deployment

Operations

SLOs, evaluation, audit, incident response

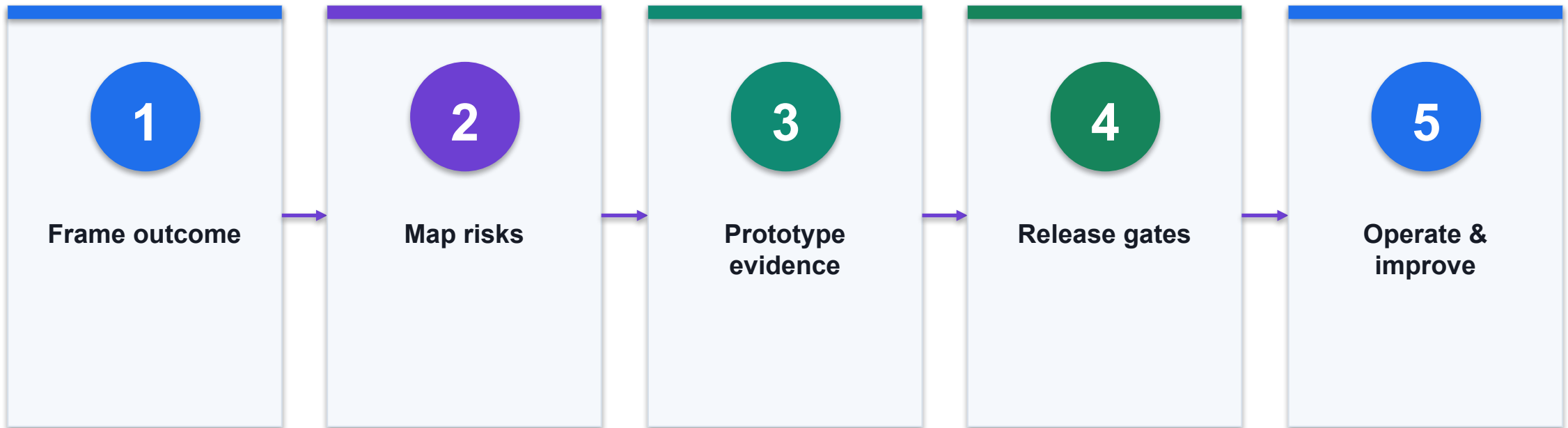
Capstone: production AI solution decision record · architecture



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Capstone: production AI solution decision record · runtime trace



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Capstone: production AI solution decision record · exact contract

Contract	Exact value / field	Engineering consequence
ADR	context, decision, alternatives, consequence	Auditable design rationale
SLO	availability, p95, quality, cost	Named owner + error budget
Evaluation	dataset, metric, threshold, version	Reproducible release gate
Runbook	signal, diagnosis, containment, recovery	Operational readiness

Capstone: production AI solution decision record · implementation excerpt

```
release_gate = {
    "groundedness": {"min": 0.82},
    "unsafe_rate": {"max": 0.01},
    "p95_ms": {"max": 2500},
    "cost_per_case_sgd": {"max": 0.08}
}
assert all(check(metric, rule) for metric, rule in
release_gate.items())
```

AUTH

Prefer Entra identity; keep secrets out of source.

CONTRACT

Pin endpoint/API/model versions and validate schema.

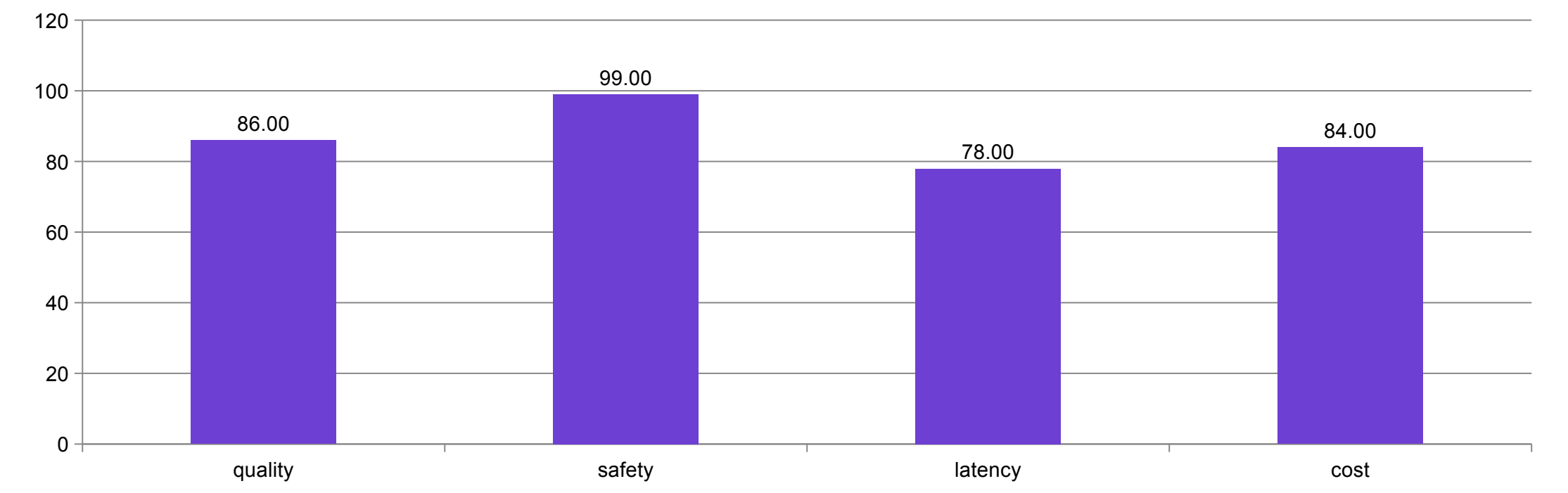
EVIDENCE

Log request IDs, deployment versions, and outcomes.

FAILURE

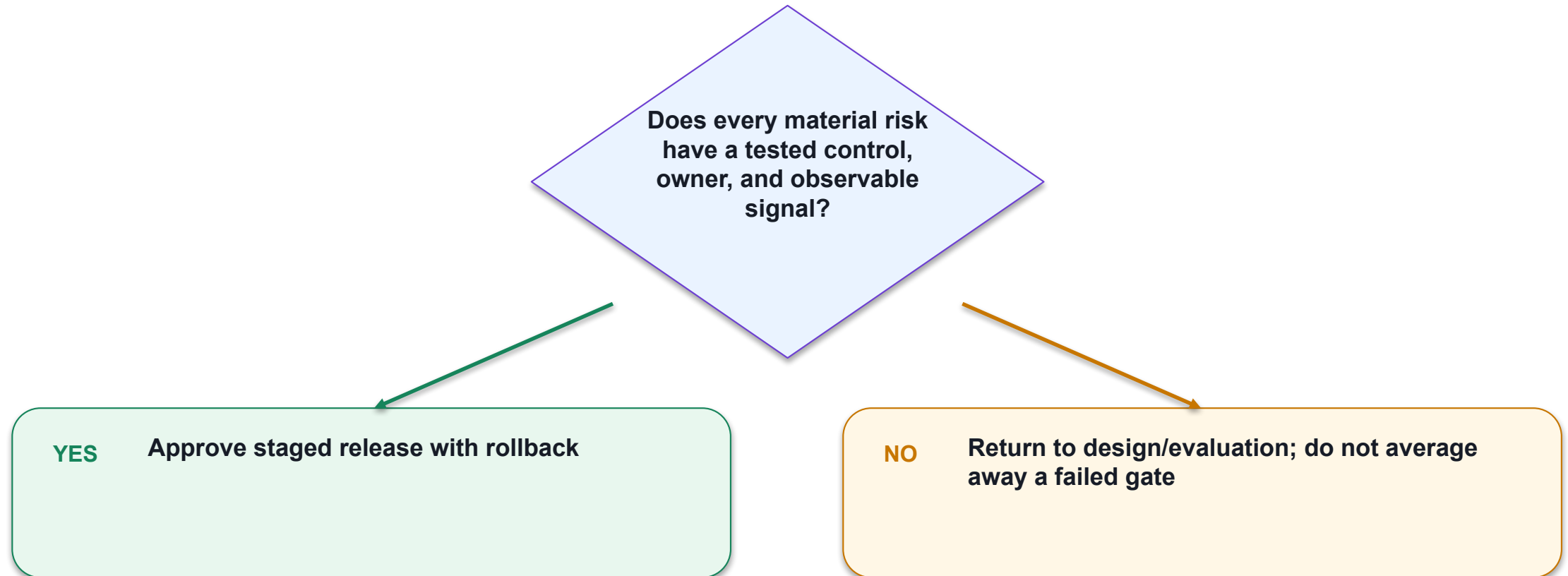
Handle auth, quota, timeout, and invalid-state paths.

Capstone: production AI solution decision record - engineering metric



WHAT THE DATA SHOWS Indicative release scorecard normalized to target: a release fails when any hard safety/quality gate fails, even if the average score looks healthy.

Capstone: production AI solution decision record · decision rule



Capstone: production AI solution decision record - failure and control

 Metric gaming LIKELY CAUSE One aggregate score CONTROL / RECOVERY Use per-risk hard gates	 No rollback LIKELY CAUSE Model/index change irreversible CONTROL / RECOVERY Blue/green assets + pinned versions	 Cost surprise LIKELY CAUSE Token/vector/skill usage unbounded CONTROL / RECOVERY Budgets, caching, quotas, alerts	 No evidence LIKELY CAUSE Decision cannot be audited CONTROL / RECOVERY Bundle ADR, tests, traces, runbook
--	--	--	--

A control is incomplete until it has a detection signal, owner, containment action, and recovery test.

Lab 10: AI-103 Capstone and Course Review

Mechanism focus: Capstone: production AI solution decision record



YOU WILL PRODUCE

You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.

Complete click-by-click procedure: [Learner Guide · Lab 10 · labs/lab-10-ai103-capstone-course-review/README.md](#)

Lab 10: acceptance evidence

PASS WHEN

Capstone evidence: architecture and data flow, ADR, threat/risk model, evaluation report, SLO dashboard, cost estimate, deployment/rollback plan, runbook, and improvement backlog.

IF IT FAILS

1. Capture request/operation ID
2. Verify identity and endpoint
3. Inspect schema/model version
4. Check quota/network/status
5. Preserve evidence before retry

Assessment bridge: submit the observable artifact and explain why the evidence proves the stated TSC ability.

TOPIC 00

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Assessment and course close

Evidence submission · assessment flow · support

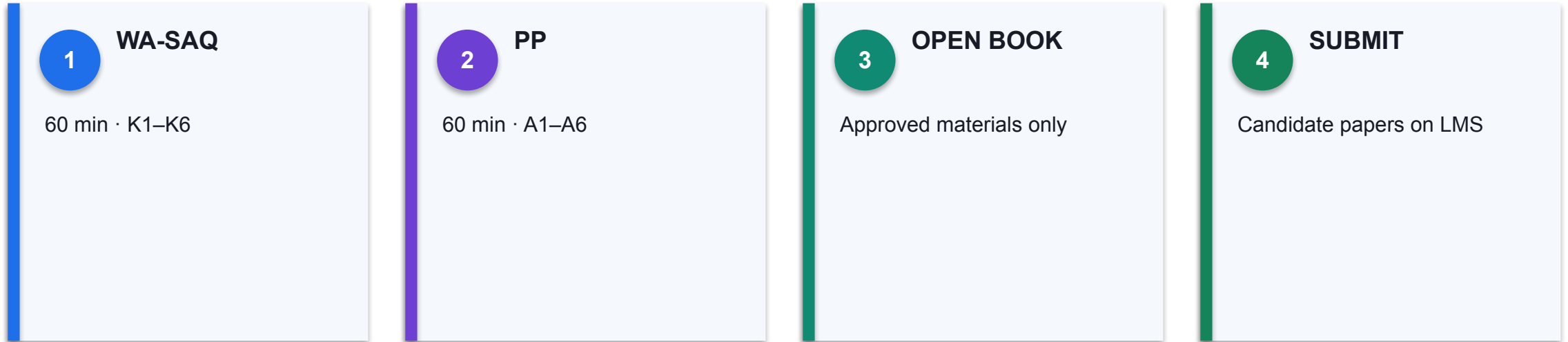
Course enquiries and support

Contact Tertiary Courses for current course details, schedules, and registration

[Tertiary Courses · Course Enquiries](#)

OPEN RESOURCE

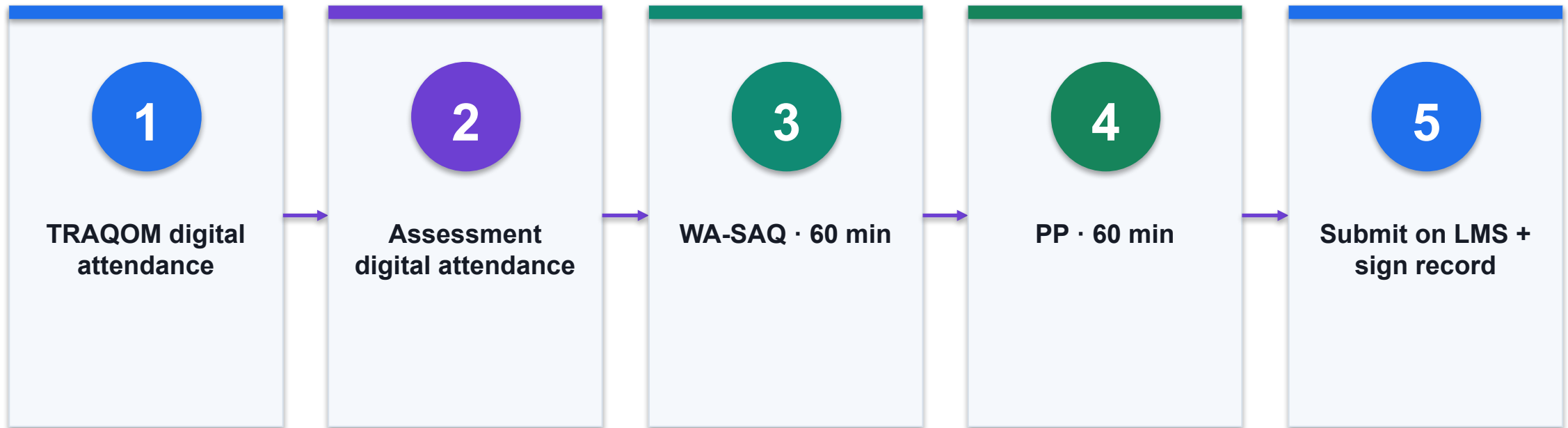
Assessment



ARCHITECTURE RULE

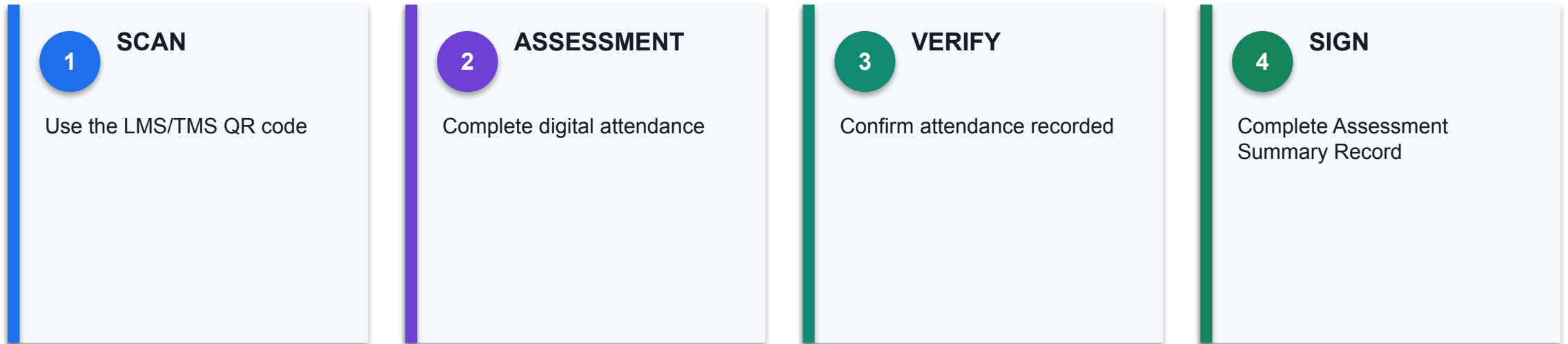
Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

Assessment Flow



Each stage emits evidence for the next stage; a missing transition or identifier is an observable contract failure.

Digital Attendance (Mandatory)



ARCHITECTURE RULE

Keep identity, data, runtime state, and evidence boundaries explicit; every connection needs an owner, contract, and observable failure signal.

TOPIC 00

00

Thank You

You can now explain, test, and operate an end-to-end Azure AI solution